

ENGG1910 Demystifying AI
Lab 5 Guidelines for Student Learning Level
Prediction:
Methods and Comparisons

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1. Objectives

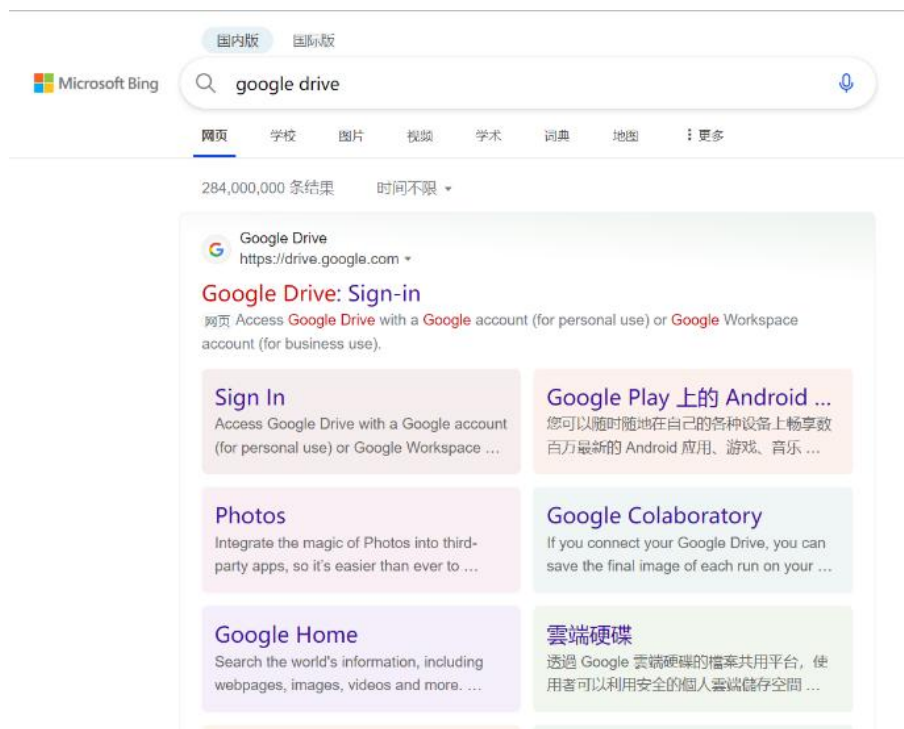
1. In this lab session, participants will be familiar with the setup of a programming platform on Colab.
2. Four learner level prediction methods will be explained (K-nearest neighbor, Support vector machine, Naïve Bayes and Back propagation neural network).
3. Participants will implement these four methods on Colab and gain some hands-on programming experience, like tuning the parameters and training the AI models.
4. The strategy for selecting the best method with optimal parameters will be shared.

2. Detailed Steps

Please kindly note that this session will be only implemented on **Colaboratory** (Colab).

2.1. Visit and log-in Google drive

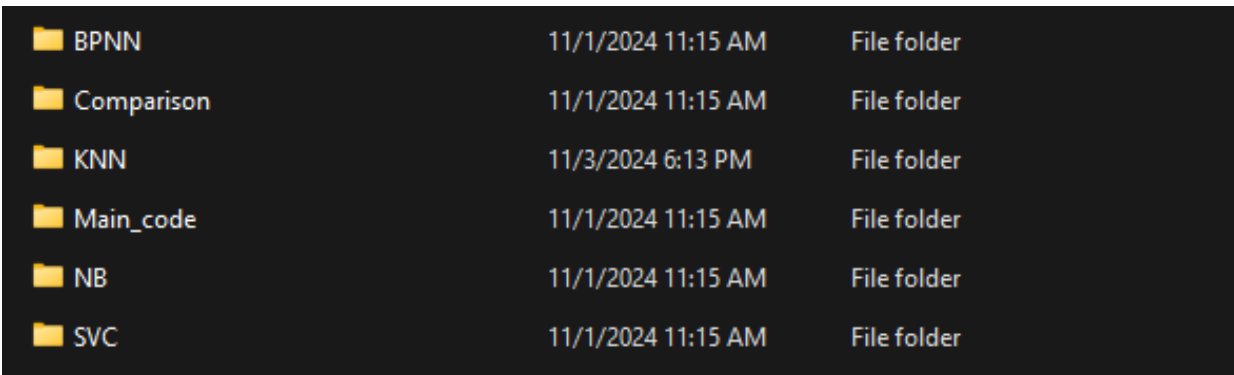
1. Search “google drive” in your browser (<https://drive.google.com/>).
2. Click the first search result.
3. Sign in with your Google account if there is one.



Picture 1: Search “google drive”

2.2. Download the files from the given google drive link

1. Download the lab materials from the folder on Blackboard:
Course Contents/Laboratories/Lab 5 Student Learning Level Prediction
2. Download all the files and documents in the folder.

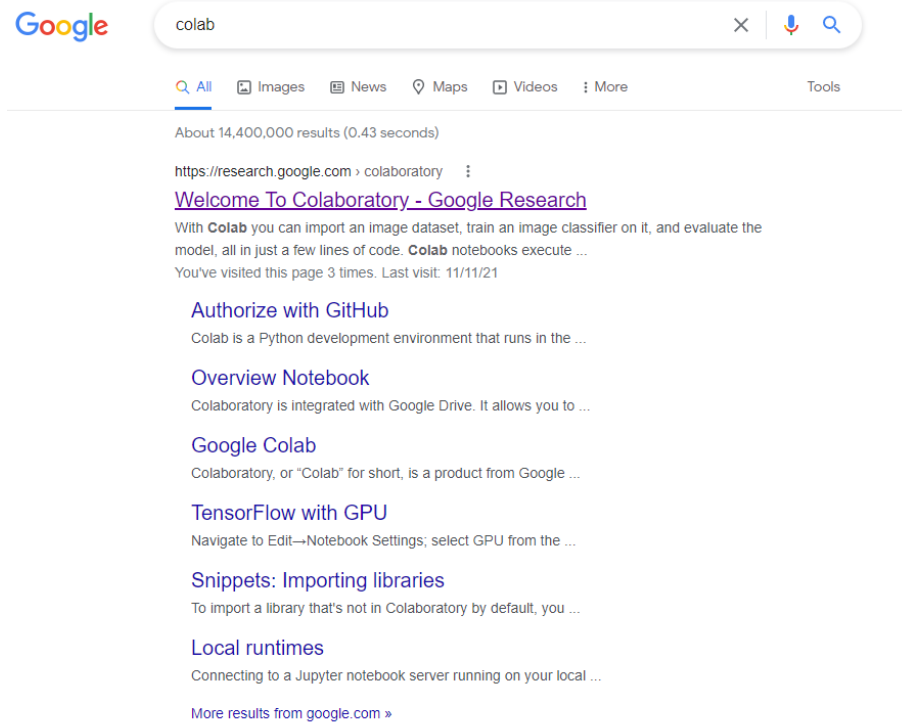
A screenshot of a Google Drive interface showing a list of folders. The background is dark grey. Each folder is represented by a yellow folder icon, its name, a timestamp, and the text 'File folder'.

BPNN	11/1/2024 11:15 AM	File folder
Comparison	11/1/2024 11:15 AM	File folder
KNN	11/3/2024 6:13 PM	File folder
Main_code	11/1/2024 11:15 AM	File folder
NB	11/1/2024 11:15 AM	File folder
SVC	11/1/2024 11:15 AM	File folder

Picture 2: Files to download

2.3. Visit and log-in Colab

1. Search “colab” in your browser (<https://drive.google.com/>).
2. Click the first search result.
3. Sign in with your Google account if there is one.

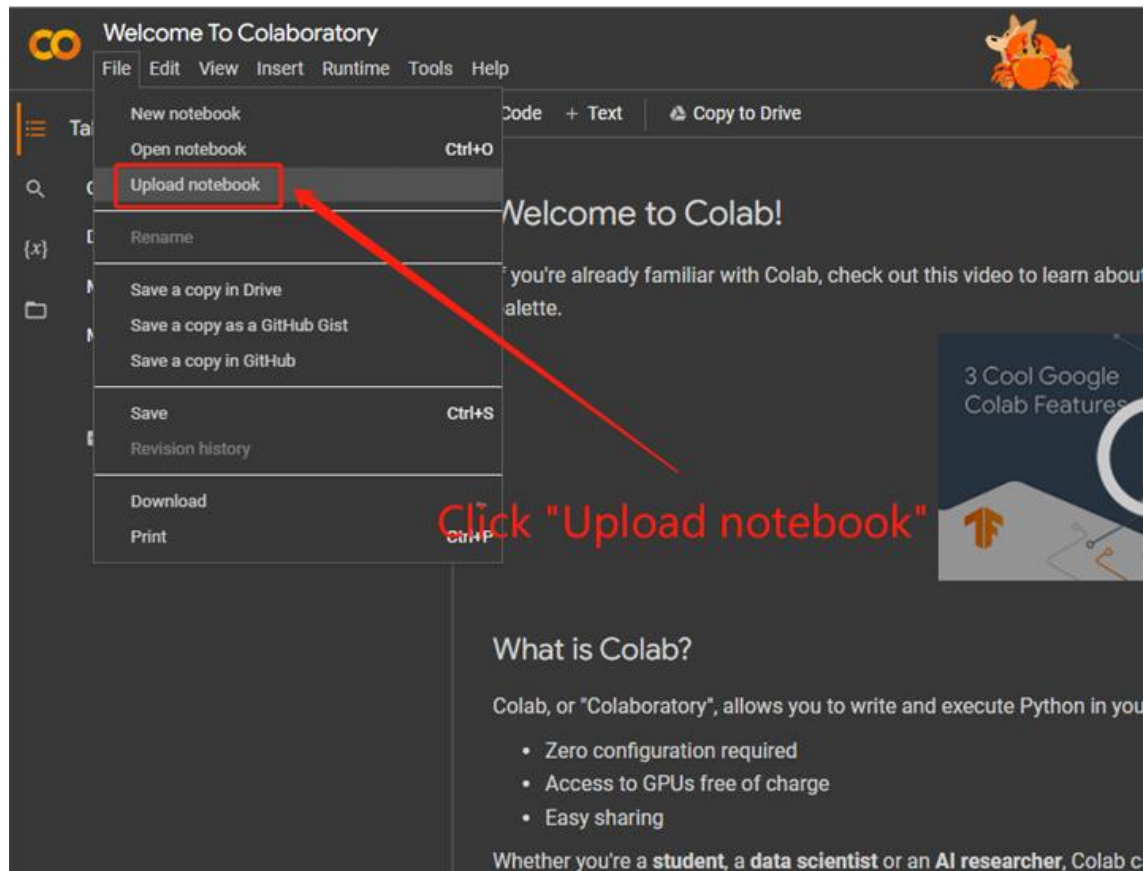


Picture 3: Log-in your account

3. KNN program

3.1. Preparations

1. Upload the “KNN.ipynb” from the file “Main_code”.



Picture 4: Upload the program

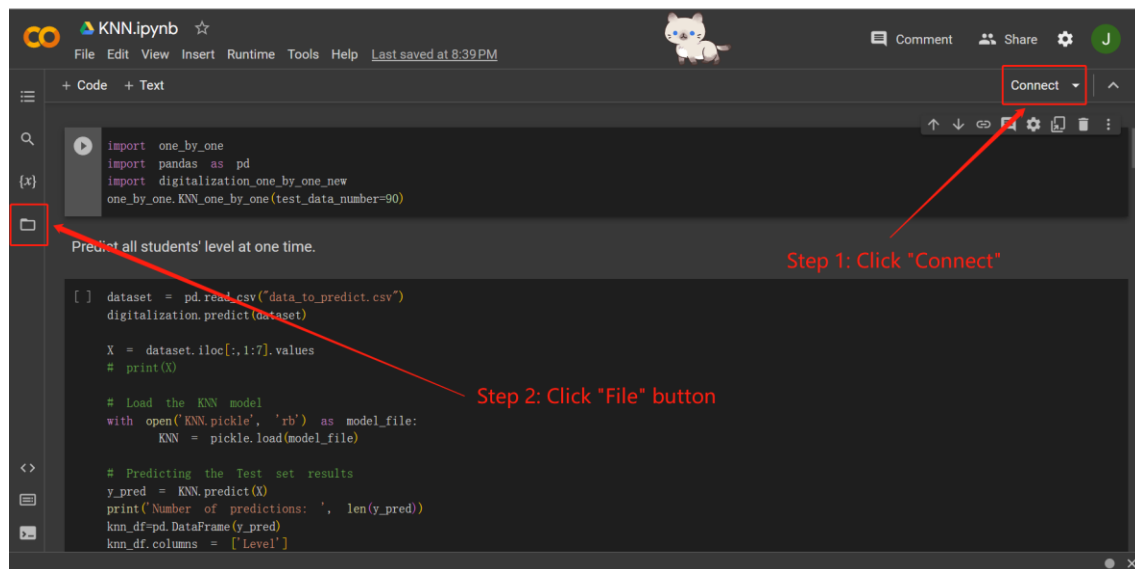
← → ▾ ↑ 📁 > Student_prediction > Main_code

名称	修改日期	类型	大小
BPNN.ipynb	2023/9/18 8:20	IPYNB 文件	15 KB
KNN.ipynb	2023/9/21 13:52	IPYNB 文件	137 KB
NB.ipynb	2023/9/18 2:13	IPYNB 文件	84 KB
SVC.ipynb	2023/9/18 1:48	IPYNB 文件	82 KB

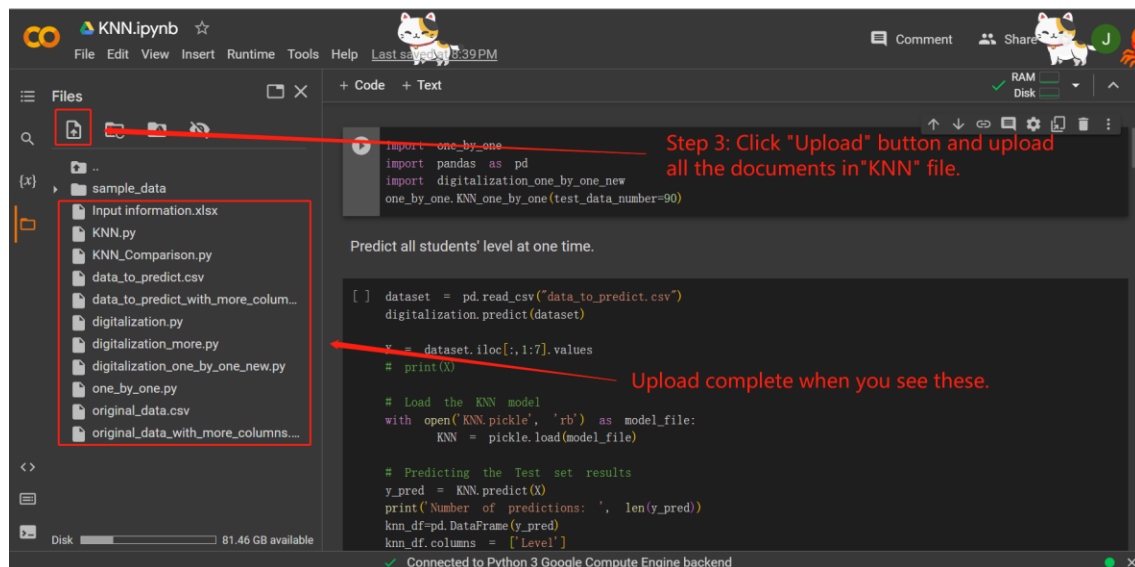
Select "KNN.ipynb"

Picture 5: Select KNN.ipynb

2. Connect the notebook to online processor.
3. Upload all the documents from "KNN" file to the notebook.

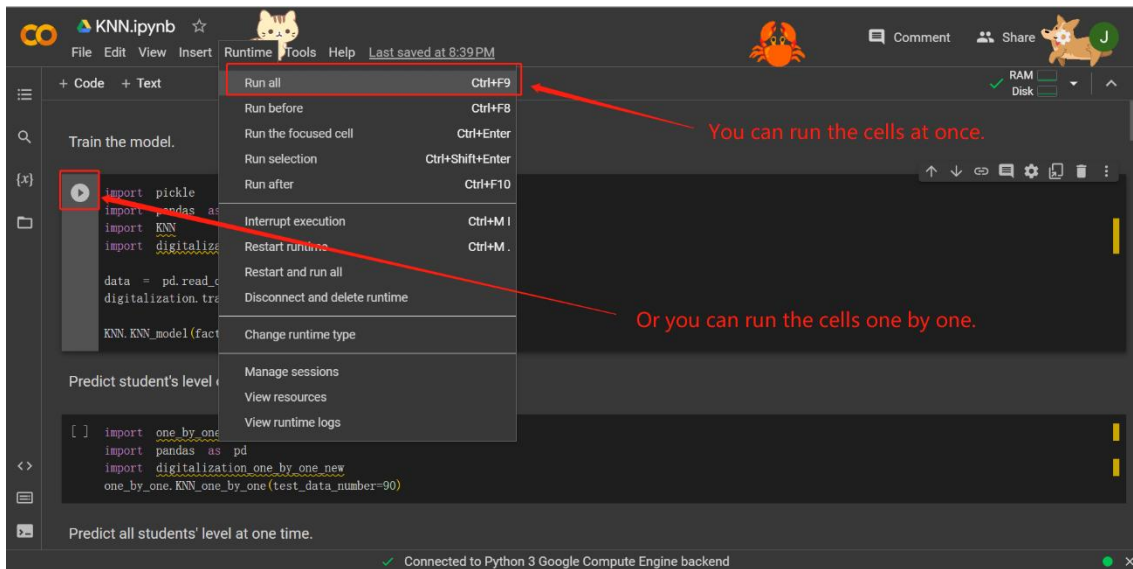


Picture 6: Connect the notebook



Picture 7: Upload our files

4. Click "Run" button and run the cells one by one or you can run them all in one click. (One by one is recommended)



Picture 8: Run the KNN program

3.2. Check the output of KNN

1. Output of the first cell shows the accuracy of our trained KNN classifier, which also represent the accuracy of final predict result.
2. Output of the second cell shows the predict level of your input student's information.
3. Output of the third cell shows the number of predictions when predict all students' level at one time.
4. Output of the fourth cell shows the relationship between different training data and the accuracy while keeping $K=3$ and test ratio of around 20%.
5. Output of the fifth cell shows the relationship between different ratio of test data and the accuracy while keeping training dataset of 545 and $K=3$.
6. Output of the sixth cell shows the relationship between different K values and the accuracy while keeping training dataset of 545 and test ratio of around 20%.
7. The final output can be found on the left, named "KNN_predict_complete.csv", after opening the file we can find that the predict level is on the rightest column.
8. After comparison, you can adjust the K value and ratio of test data like in the picture below. (Since training data is generally the more the better so there's no need to change)

```

Help All changes saved
+ Code + Text
Train the model.
import pickle
import pandas as pd
import KNN
import digitalization
data = pd.read_csv('original_data.csv')
digitalization.train(data)
KNN_KNN_model(factor_number=7, k=3, test_data_number=90)
Accuracy of the KNN Classification is: 0.9644444444444444
Predict student's level one by one based on input value.
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
0s completed at 8:58 PM

```

Here shows the accuracy of the algorithm

Picture 9: First output

```

Predict student's level one by one based on input value.
import one_by_one
import pandas as pd
import digitalization_one_by_one_new
one_by_one.KNN_one_by_one(test_data_number=90)
Please enter the value of k: 3
Please enter Admission_Channel: JUPAS
Please enter Secondary_School: ST PAULS COLLEGE
Please enter Exam: HKDSE
Please enter Math_Background: HKDSE:5*
Please enter MATH1510: A
Please enter ENGG1130: A-
predict level is ['A']
Predict all students' levels at one time.

```

The predicted level is A

Picture 10: Second output

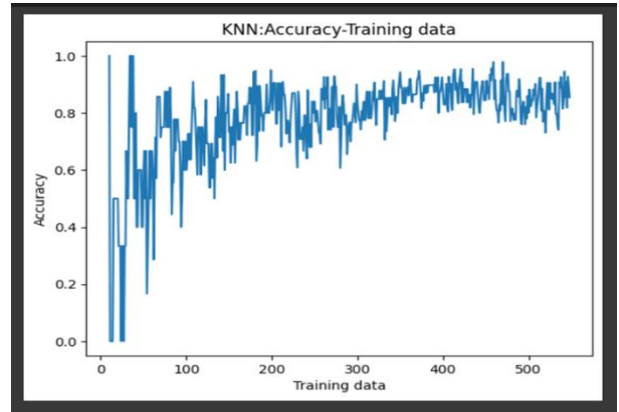
```

Help All changes saved
+ Code + Text
Predict all students' level at one time.
dataset = pd.read_csv('data_to_predict.csv')
digitalization.predict(dataset)
X = dataset.iloc[:, 1:11].values
y = dataset.iloc[:, 11].values
# Load the KNN model
with open('KNN_pickle', 'rb') as model_file:
    KNN = pickle.load(model_file)
# Predicting the test set results
y_pred = KNN.predict(X)
print('Number of predictions: ', len(y_pred))
knn_df=pd.DataFrame(y_pred)
knn_df.columns = ['level']
pd.io.formats.printer.to_csv(knn_df, index=False)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('KNN_pred_level.csv')
file = f1.f2
train = pd.concat(file, axis=1)
train.to_csv('KNN_predict_complete' + '.csv', index=False, sep=',')
Number of predictions: 545
0s completed at 9:16 PM

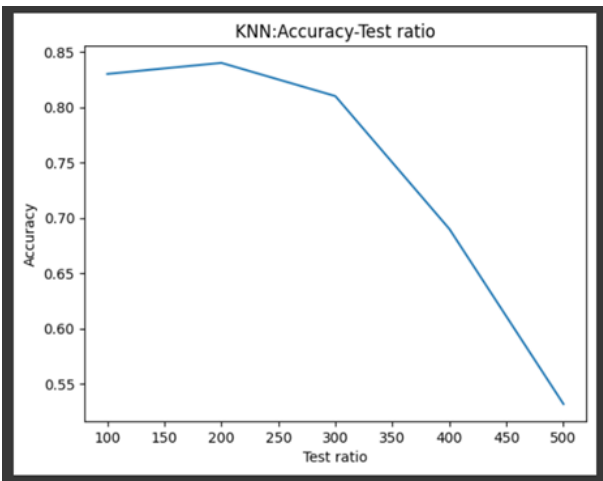
```

Here shows the number of predictions

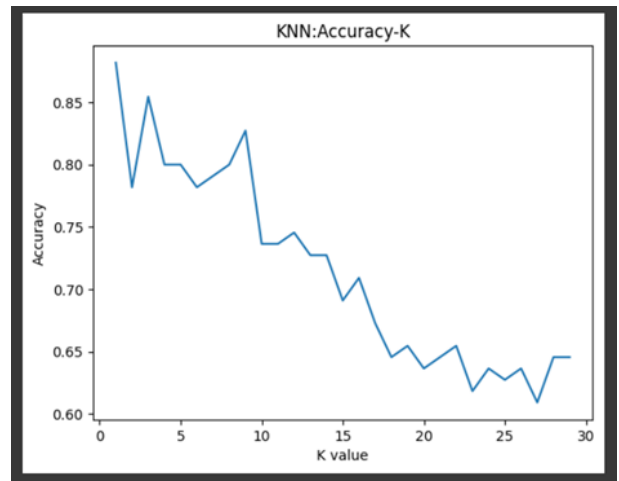
Picture 11: Third output



Picture 12: Fourth output



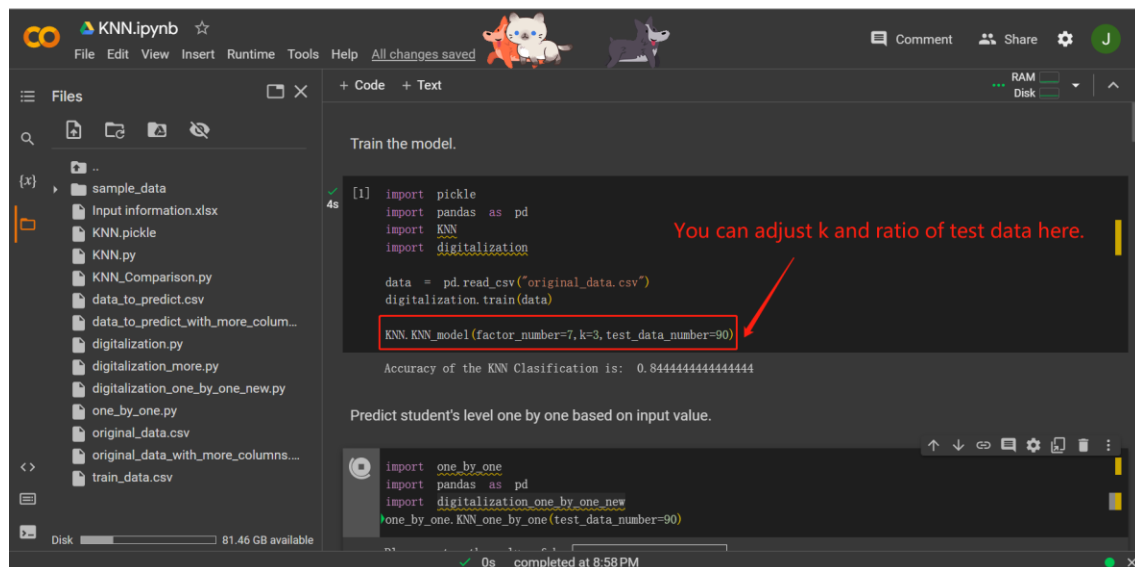
Picture 13: Fifth output



Picture 14: Sixth output



Picture 15: Outcome

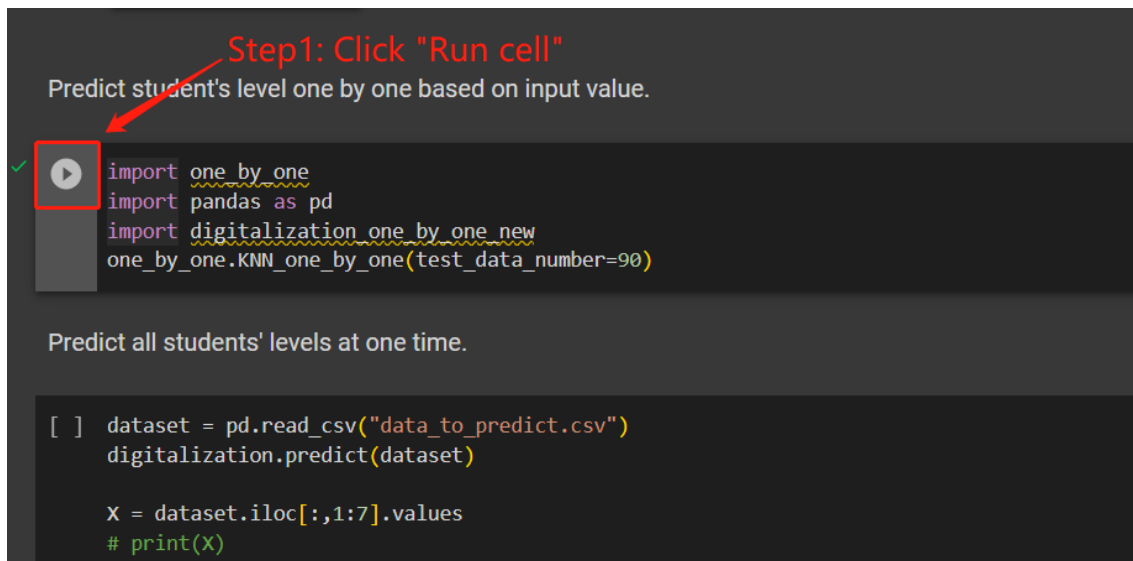


Picture 16: Adjust the parameters

3.3. Predict Student's level one by one

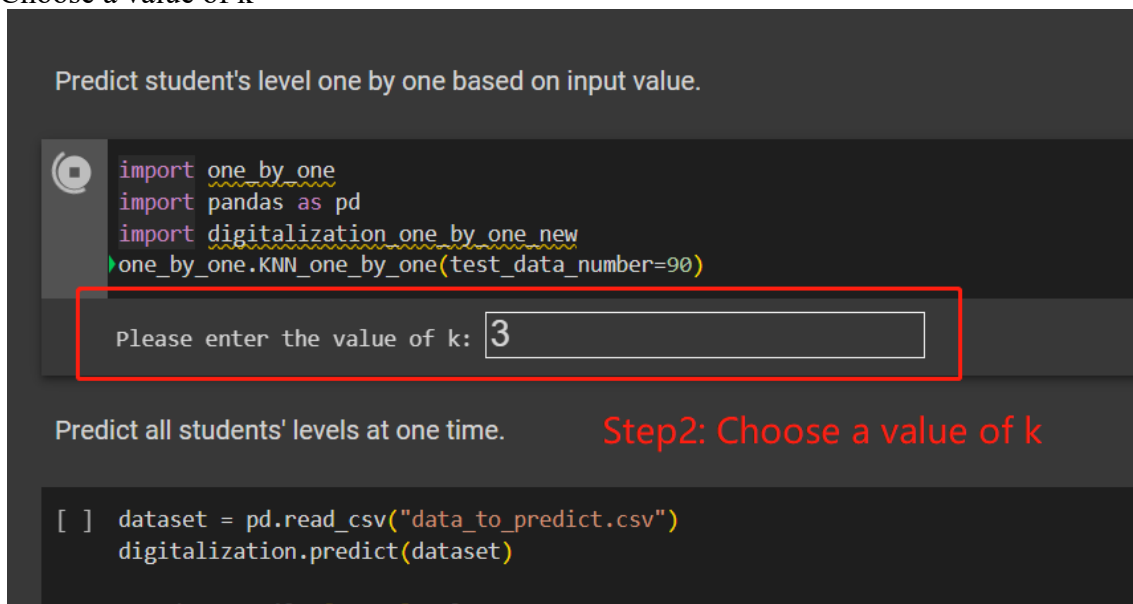
In this task, you can input one student's information (Admission channel, secondary school, exam type, math background, grade of MATH1510, and grade of ENGG1130), and get the predicted the grade of ENGG2020 for this student.

1. Let's run the second cell of the KNN code. Firstly, click "run the code".



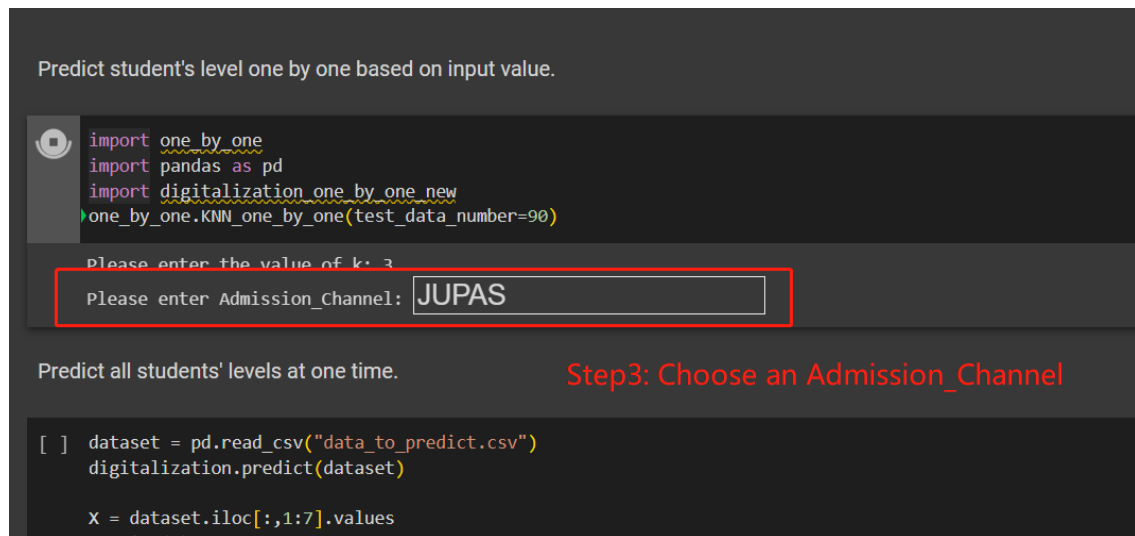
Picture 17: Run the code

2. Choose a value of k



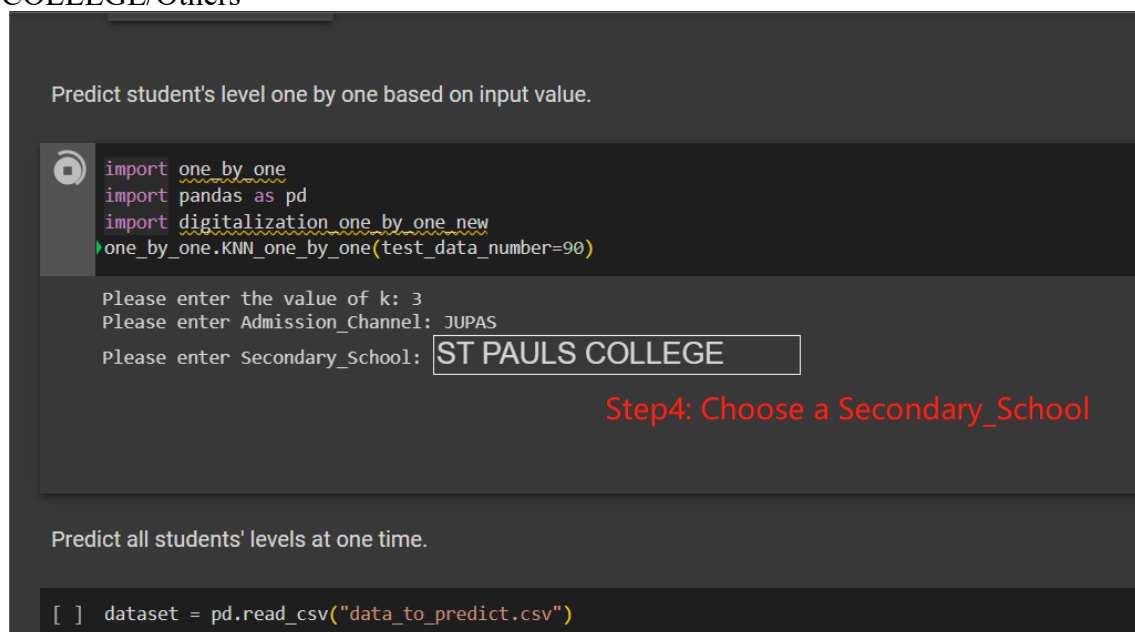
Picture 18: Choose k

3. Enter the student's admission channel, you can choose a channel from JUPAS/Mainland Fee-paying/Mainland Scholarship (Tuition)/International/JUPAS Appeal-in.



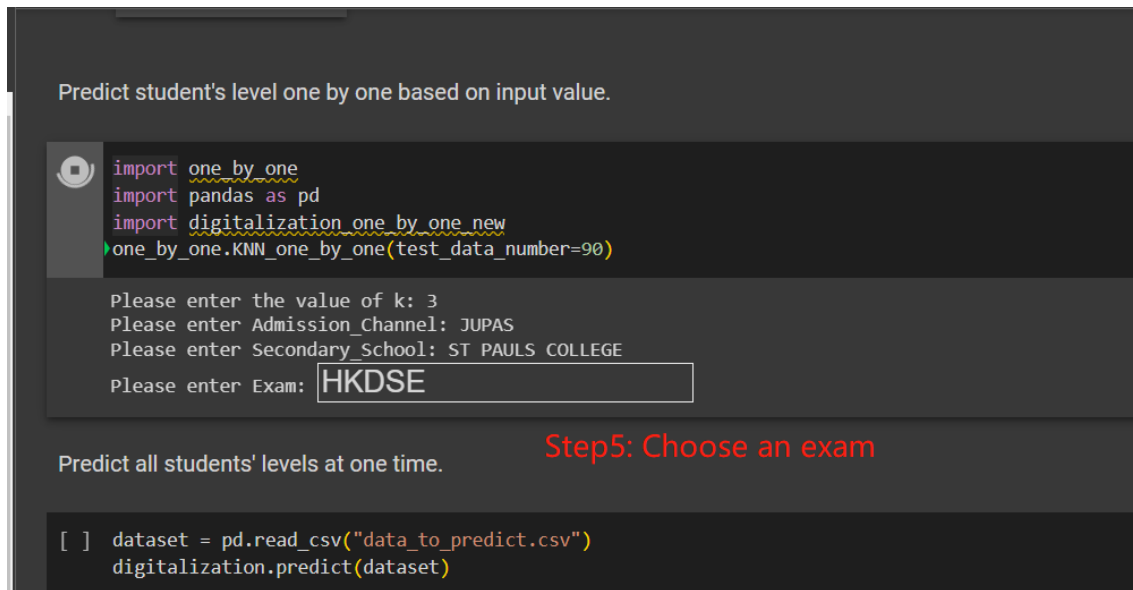
Picture 19: Choose Admission_Channel

4. Enter the student's secondary school, you can choose a school from ST PAULS COLLEGE/CNEC CHRISTIAN COLLEGE/LA SALLE COLLEGE/HKRSS TAI PO SECONDARY SCHOOL/KAU YAN COLLEGE/ SHA TIN METHODIST COLLEGE/Others



Picture 20: Choose Secondary School

5. Enter the student's exam type, you can choose from HKDSE/Mainland Gaokao/Mainland JEE/Others

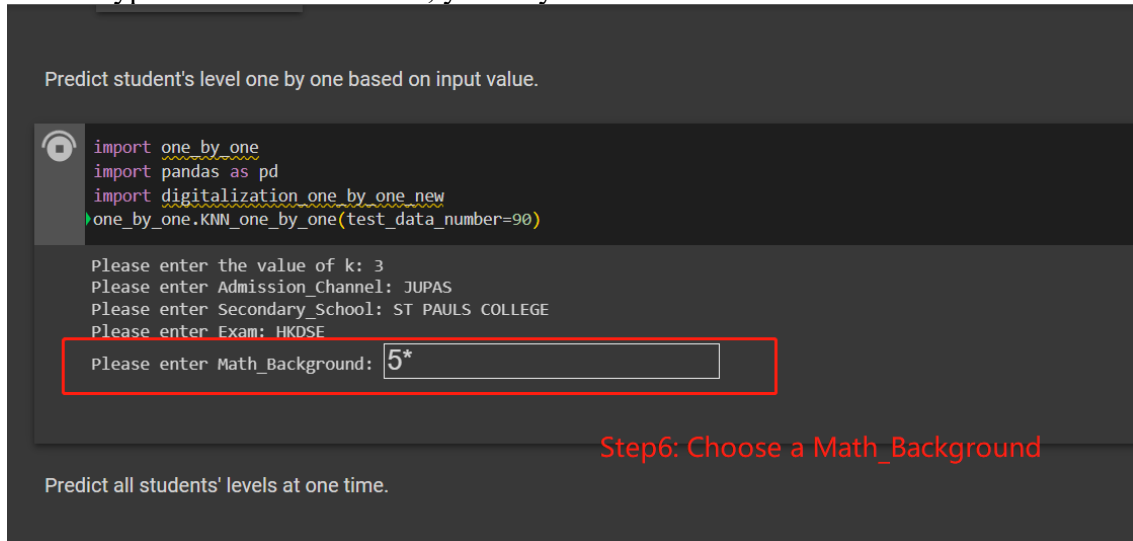


Picture 21: Choose exam type

6. Enter the student's math background.

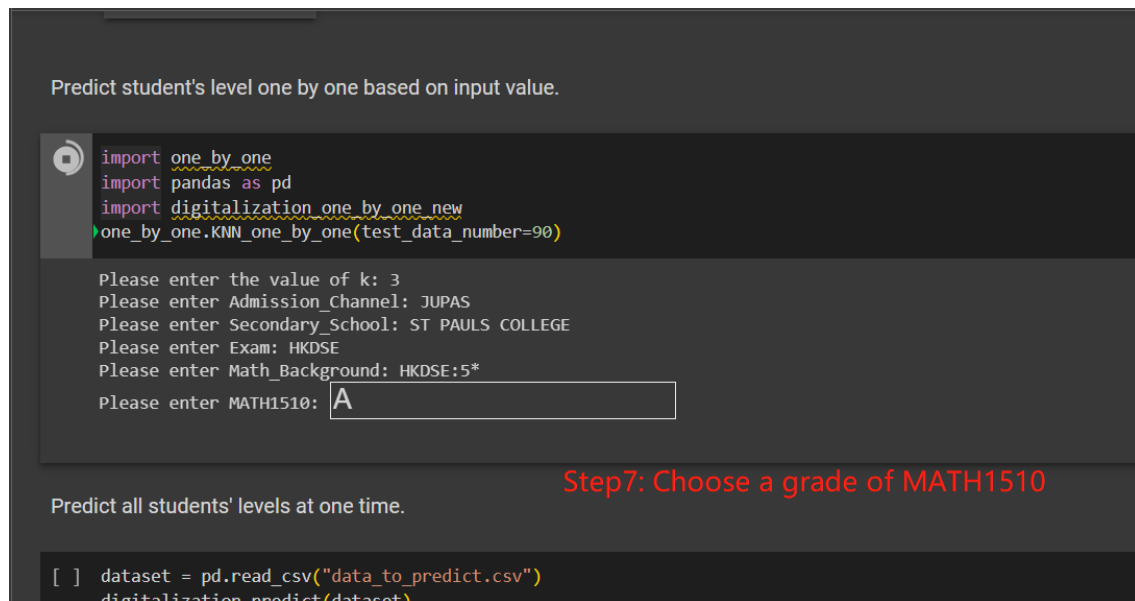
If the exam type is HKDSE, you may choose from 5*/5*/5/4/3/2/1.

If the exam type is Mainland Gaokao, you may choose from >=140/>=130/>=120/>=110/<110.



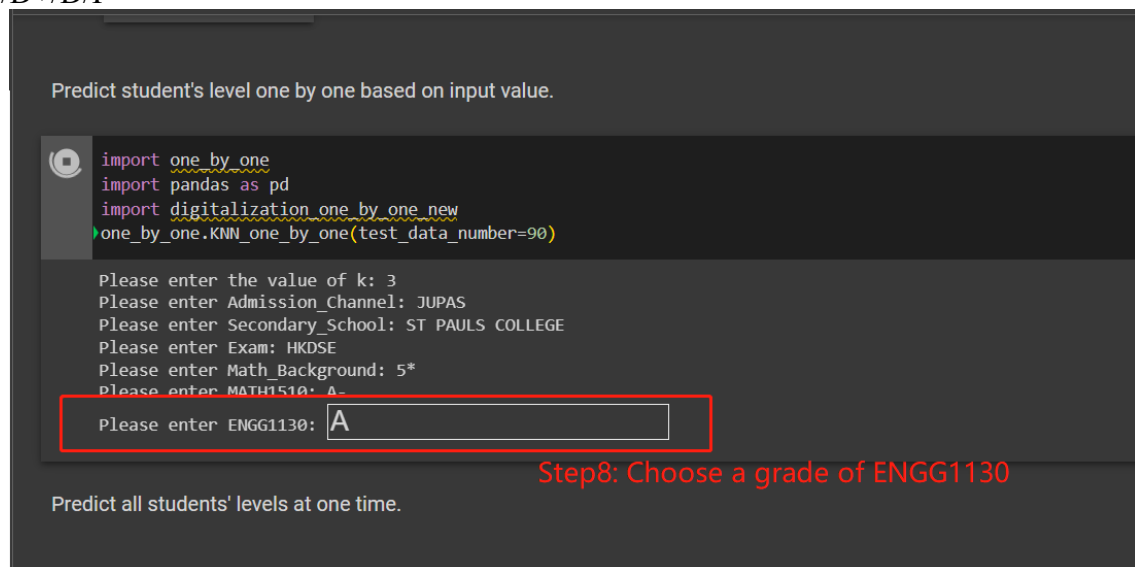
Picture 22: Choose Math_Background

7. Enter the student's grade of MATH1510, you can choose from A/A-/B+/B/B-/C+/C/C-/D+/D



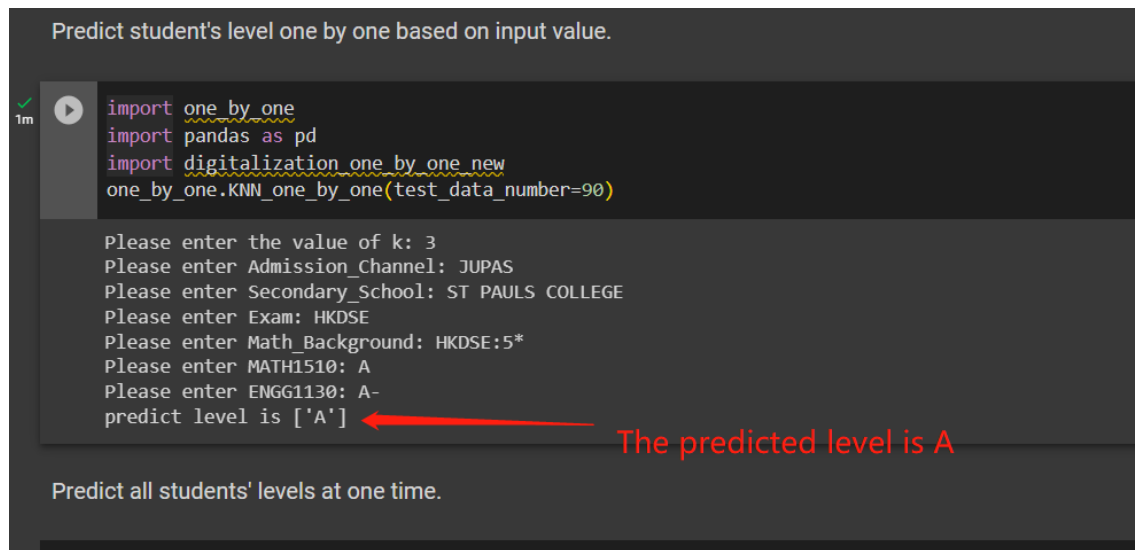
Picture 23: Choose grade of MATH1510

8. Enter the student's grade of ENGG1130, you can choose from A/A-/B+/B/B-/C+/C/C-/D+/D/F



Picture 24: Choose grade of ENGG1130

9. Then you can get the predicted math level of this student.

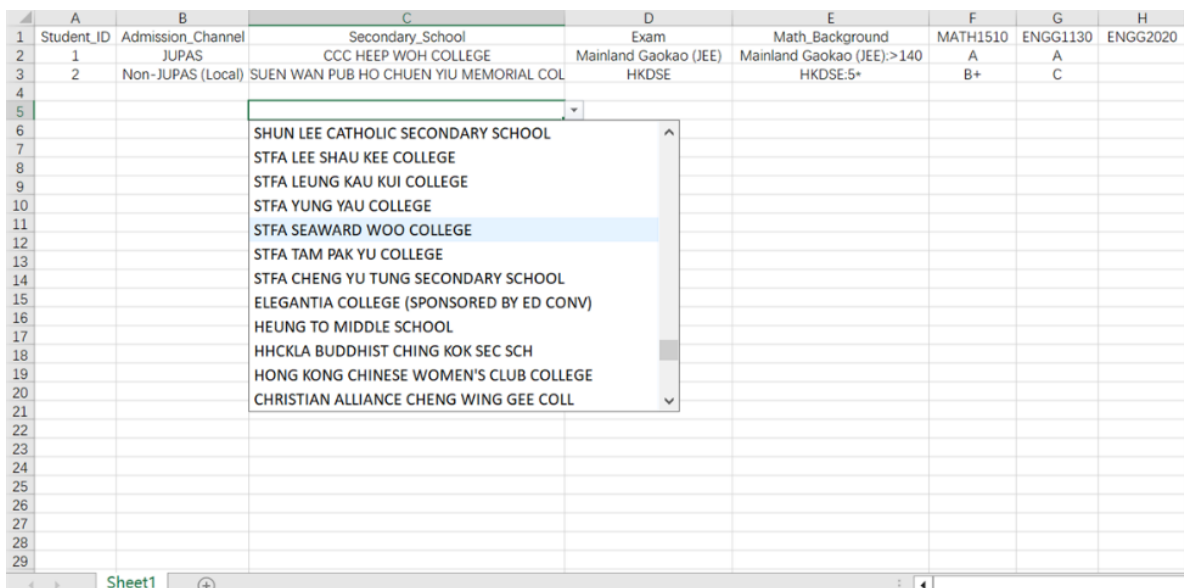


Picture 25: Predicted level

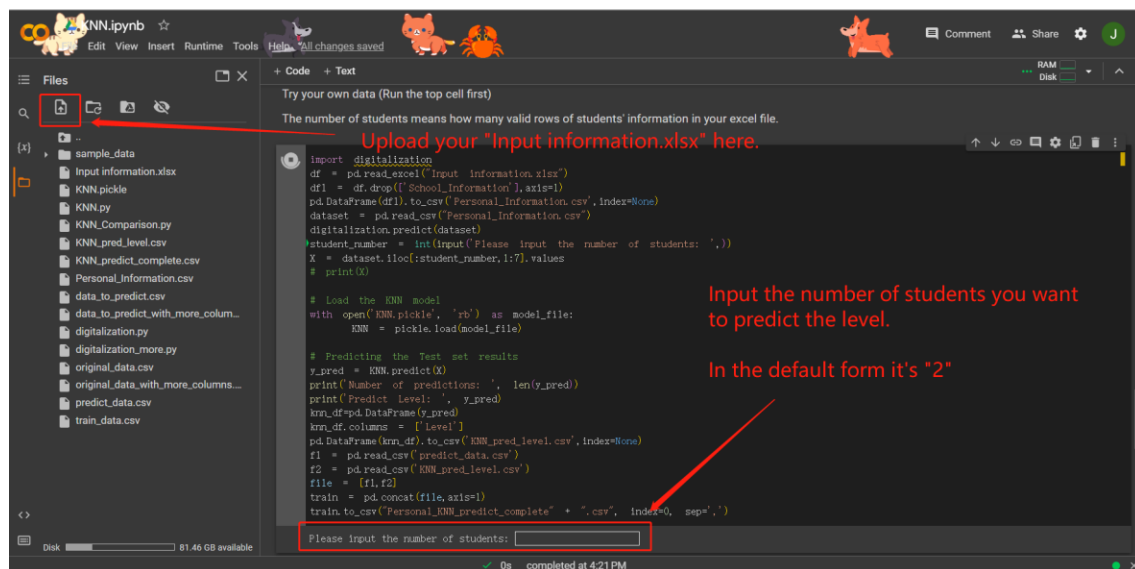
3.4. Want to try your own data?

Follow the steps below:

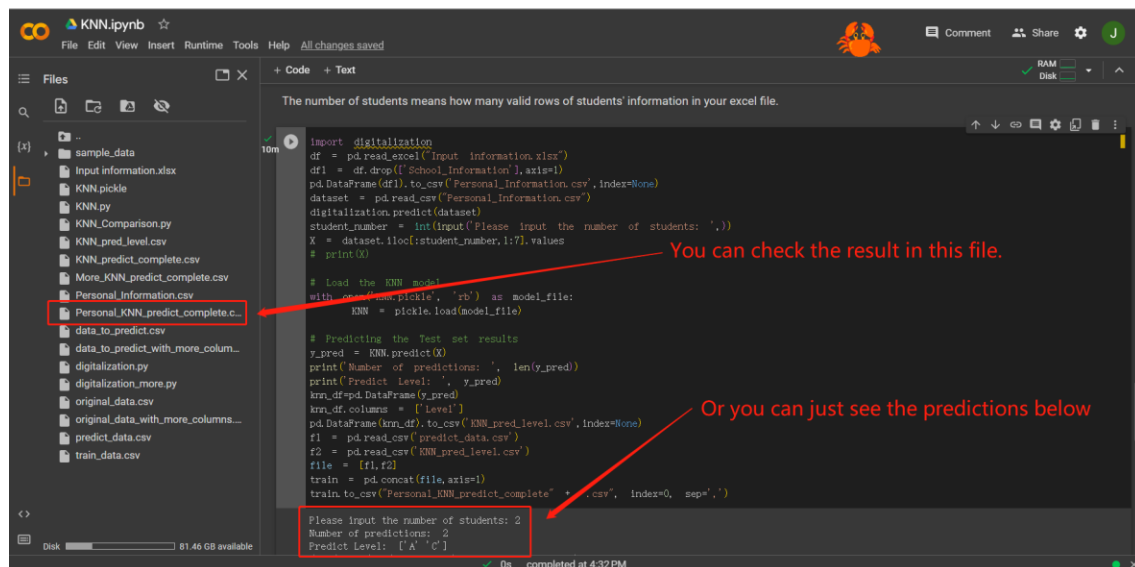
1. First, open the “Input information.xlsx” from the “KNN” file we have already downloaded, fill in your personal information. Notice that in “Exam” and “Math_Background” column, only Gaokao and HKDSE are accepted.
2. Upload the file to KNN program and run the seventh cell.
3. Then pick up the “Personal_KNN_predict_complete.csv” file to check the prediction of grades.



Picture 26: Input your information



Picture 27: Run the seventh cell



Picture 28: Two ways to check the predictions

3.5. Consider additional factors

Follow the steps below:

1. We can take a look at "original_data_with_more_columns.csv" in "KNN" file.
2. After training the model with the dataset in previous slide, you can make a csv document contains the students' information you want to predict their level ("data_to_predict_with_more_columns.csv").

Original No.

We add two more courses as factors for example

	A	B	C	D	E	F	G	H	I	J
1	Original No.	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
2	17 JUPAS		TSUEN WAN PUB HO CHUEN YIU MEMORIAL COLL	HKDSE	HKDSE:5	B+	B+	A-	B	B+
3	20 JUPAS		CHRISTIAN ALLIANCE S C CHAN MEM COLL	HKDSE	HKDSE:5*	A-	B	B	B+	B
4	25 JUPAS		MUNSANG COLLEGE (HONG KONG ISLAND)	HKDSE	HKDSE:4	B	A-	A-	B+	A
5	29 JUPAS		PO LEUNG KUK LAWS FOUNDATION COLLEGE	HKDSE	HKDSE:4	B+	B-	B+	A-	B-
6	36 JUPAS		STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5	B-	B+	B+	B	B+
7	37 JUPAS		METHODIST CHURCH HK WESLEY COLLEGE	HKDSE	HKDSE:5	D	D+	B+	B+	D
8	50 JUPAS		Others	HKDSE	HKDSE:5	B-	C+	D	D+	C
9	55 JUPAS		KWOK TAK SENG CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:5	B	B	B	C+	B
10	62 JUPAS		YWCA HIOE TJO YOENG COLLEGE	HKDSE	HKDSE:4	B	B+	B-	B	B+
11	82 JUPAS		CCC CHUEN YUEN COLLEGE	HKDSE	HKDSE:5	B	F	B	B+	F
12	88 JUPAS		HKMA K S LO COLLEGE	HKDSE	HKDSE:4	C	C+	C-	F	C
13	92 JUPAS		LIONS COLLEGE	HKDSE	HKDSE:4	B	C-	C+	C+	C
14	94 JUPAS		YCH LIM POR YEN SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	A-	C-	C-	A
15	98 JUPAS		HK SKH BISHOP HALL SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	B+	A-	A-	B+
16	101 JUPAS		CHRISTIAN ALLIANCE S W CHAN MEM COLL	HKDSE	HKDSE:5	B-	B-	B+	B+	B-
17	109 JUPAS		VALTORTA COLLEGE	HKDSE	HKDSE:5**	A-	A	B-	B+	A
18	119 JUPAS		AOG HEBRON SECONDARY SCHOOL	HKDSE	HKDSE:5	B-	A-	A	A	A
19	129 JUPAS		PUI CHING MIDDLE SCHOOL	HKDSE	HKDSE:5**	A-	A	A-	A-	A
20	130 JUPAS		TWGH LI KA SHING COLLEGE	HKDSE	HKDSE:5	C+	C-	A	A	C
21	142 JUPAS		TANG KING PO SCHOOL	HKDSE	HKDSE:4	B+	B-	C-	C+	B-
22	144 JUPAS		PLK MRS MA KAM MING CHEUNG FOOK SIEN COL	HKDSE	HKDSE:4	B+	B-	B-	B-	B-
23	147 JUPAS	Appeal-in	STFA YUNG YAU COLLEGE	HKDSE	HKDSE:5	B-	C	B-	B-	C
24	168 JUPAS		FUKIEN SECONDARY SCHOOL	HKDSE	HKDSE:5*	A-	B+	C	C	B+
25	172 JUPAS		SKH LAM WOO MEMORIAL SECONDARY SCHOOL	HKDSE	HKDSE:4	B	C+	B+	B+	C

Picture 29: original_data_with_more_columns.csv

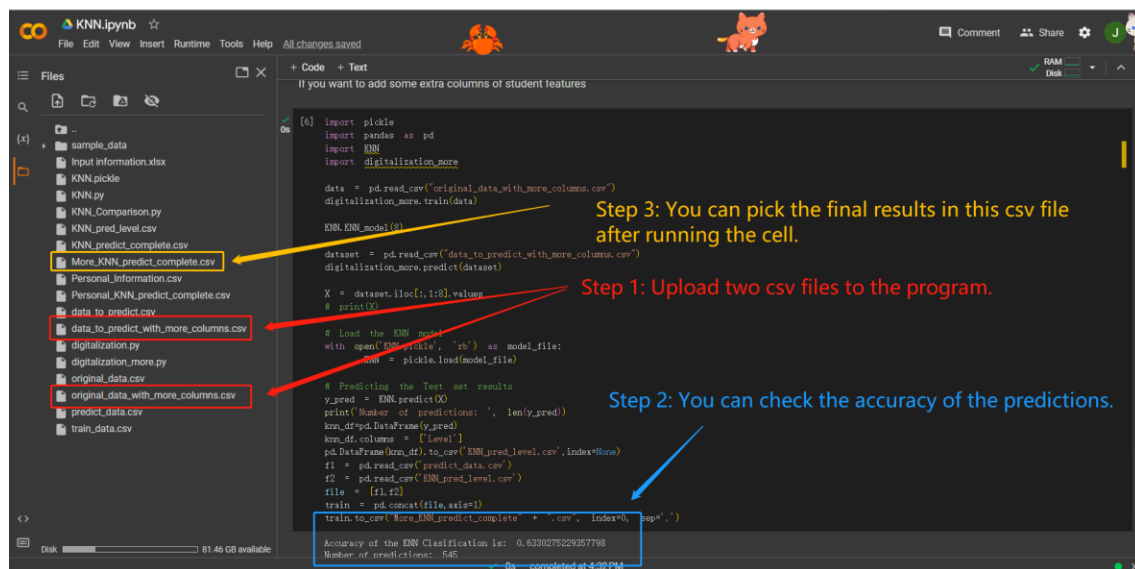
Student_ID

After running the cell, the level of these students will be added

	A	B	C	D	E	F	G	H	I	J
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
2	627 JUPAS		CHAN SUI KI (LA SALLE) COLLEGE	HKDSE	HKDSE:5*	A-	A-	B-	B-	
3	882 JUPAS		CHRISTIAN ALLIANCE CHENG WING GEE COLL	HKDSE	HKDSE:4	D	F	B	B+	
4	801 JUPAS		MA ON SHAN TSUNG TSIN SECONDARY SCHOOL	HKDSE	HKDSE:5*	B-	B	C	C+	
5	1087 Mainland Fee-paying		Others	Mainland	(Mainland Gaokao (JEE):<110	B-	B	B-	C+	
6	224 JUPAS		ELEGANTIA COLLEGE (SPONSORED BY ED CONV)	HKDSE	HKDSE:5*	B-	F	B-	B-	
7	240 JUPAS		SKH BISHOP MOK SAU TSENG SEC SCH	HKDSE	HKDSE:5*	A-	A-	B-	B+	
8	911 JUPAS		Others	HKDSE	HKDSE:5**	A	A	B-	B+	
9	917 JUPAS		DIOCESAN BOYS SCHOOL	HKDSE	HKDSE:4	D	B-	B	B+	
10	624 Mainland		Others	Mainland	(Mainland Gaokao (JEE):120-129	A-	A	B	A	
11	724 JUPAS		SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B	B-	B+	
12	546 JUPAS		KWUN TONG MARYKNOLL COLLEGE	HKDSE	HKDSE:5*	A	A	B	B+	
13	410 Mainland		Others	Mainland	(Mainland Gaokao (JEE):130-139	B+	B	A-	B	
14	492 JUPAS		Others	HKDSE	HKDSE:5	B-	A-	B	B-	
15	986 JUPAS		Others	HKDSE	HKDSE:4	C+	B+	A-	B+	
16	1046 JUPAS		HELEN LIANG MEMORIAL SEC SCH (SHATIN)	HKDSE	HKDSE:5	A-	B+	B	B	
17	272 JUPAS		HEUNG TO MIDDLE SCHOOL	HKDSE	HKDSE:4	B-	C-	A-	A-	
18	803 JUPAS		Others	HKDSE	HKDSE:4	B-	B+	B	B-	
19	383 JUPAS		LAI KING CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:4	C-	B	B-	B	
20	881 JUPAS		YAN OI TONG TIN KA PING SECONDARY SCHOOL	HKDSE	HKDSE:4	B-	C+	B+	B	
21	932 JUPAS		SKH TANG SHIU KIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B+	B-	A-	
22	120 Mainland		Others	Mainland	(Mainland Gaokao (JEE):120-129	A	A	A-	A	
23	732 JUPAS		HKSYC&JA WONG TAI SHAN MEM COLL	HKDSE	HKDSE:4	C	B	B+	B+	
24	474 JUPAS		SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	A	B-	C-	
25	776 JUPAS		STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5*	A	A	B-	B-	

Picture 30: data_to_predict_with_more_columns.csv

- Then upload two csv files to the program.
- Run the eighth cell then you can check the accuracy and pick the final results in the “More_KNN_predict_complete.csv”.

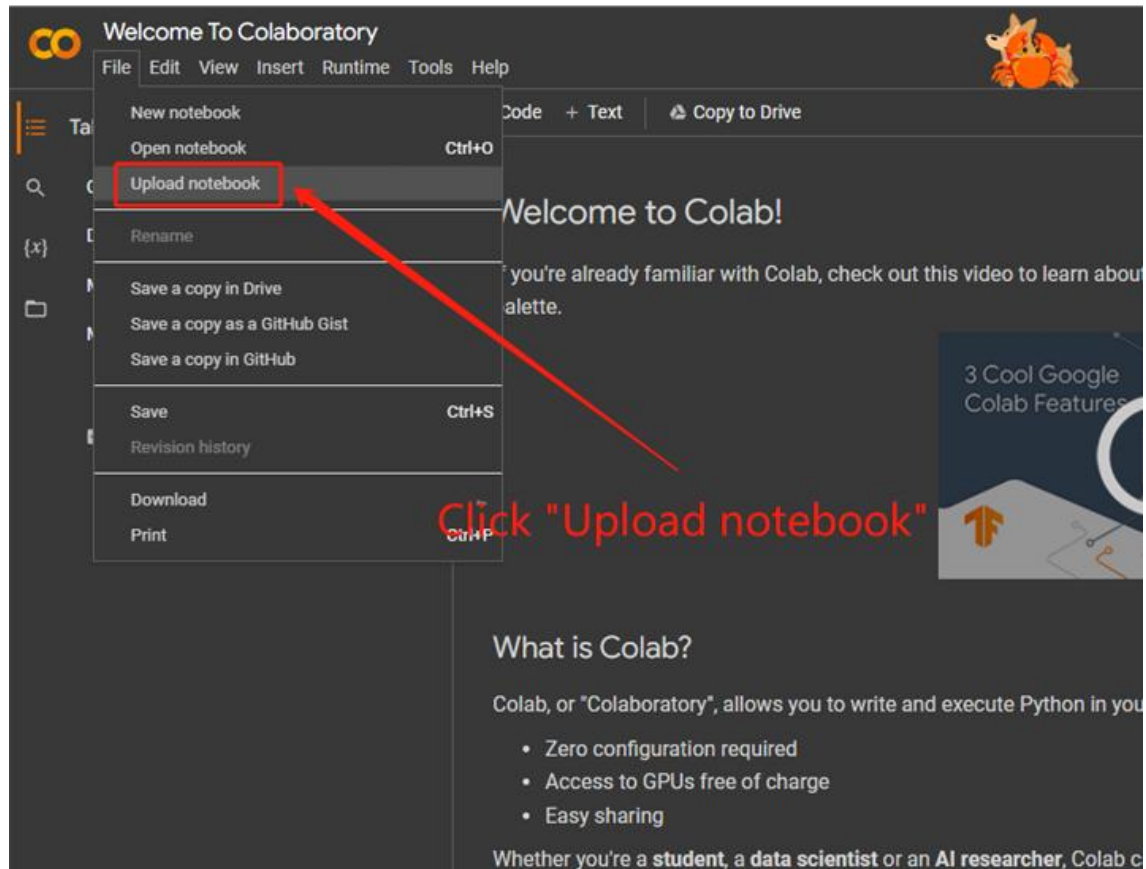


Picture 31: Run the eighth cell and get your results

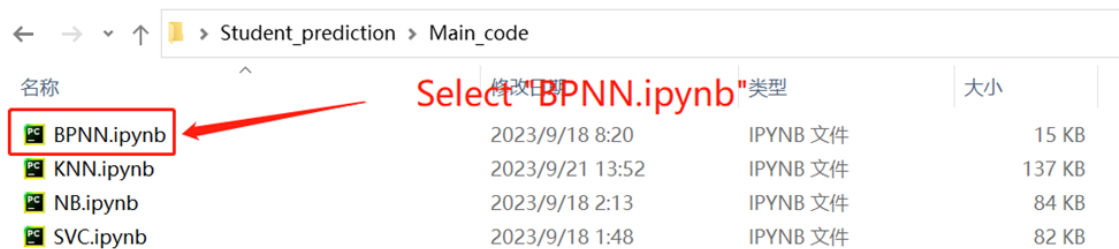
4. BPNN program

4.1. Preparations

1. Upload the “BPNN.ipynb” from the file “Main_code”.

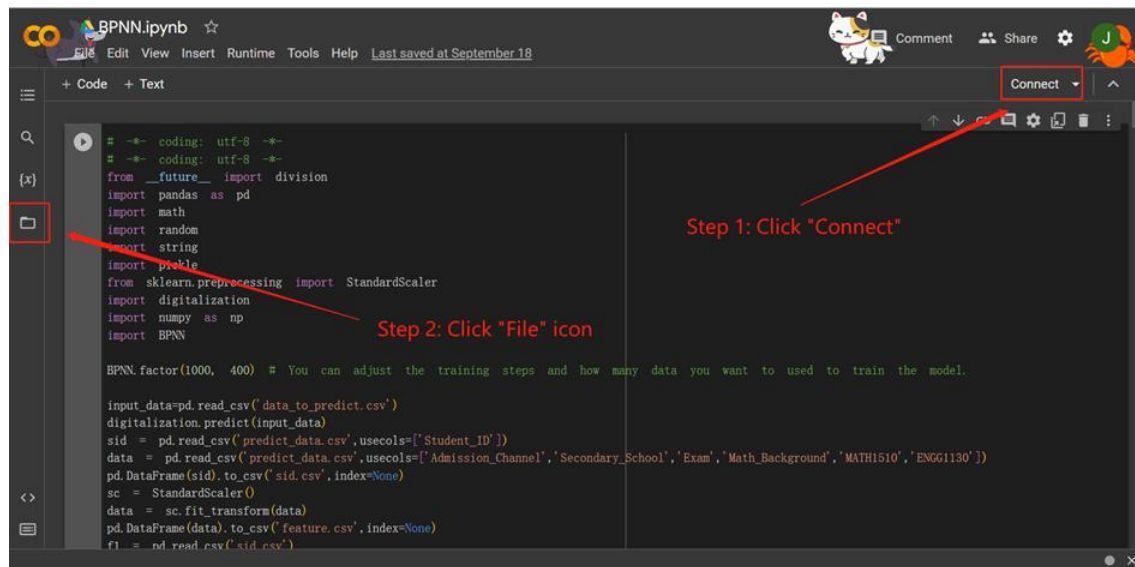


Picture 32: Upload the program

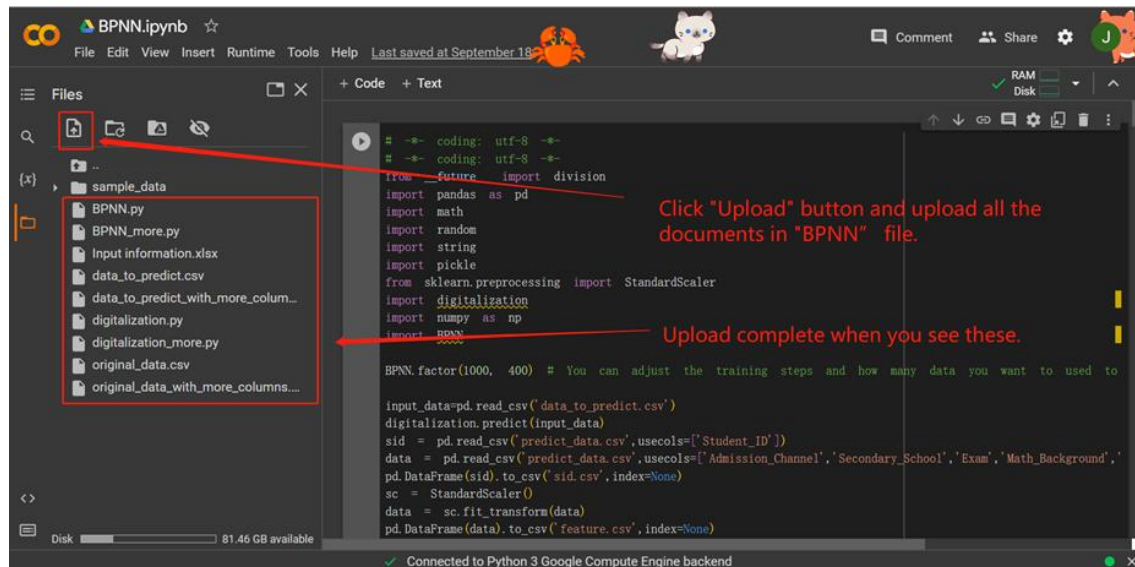


Picture 33: Select BPNN.ipynb

2. Connect the notebook to online processor.
3. Upload all the documents from “BPNN” file to the notebook.

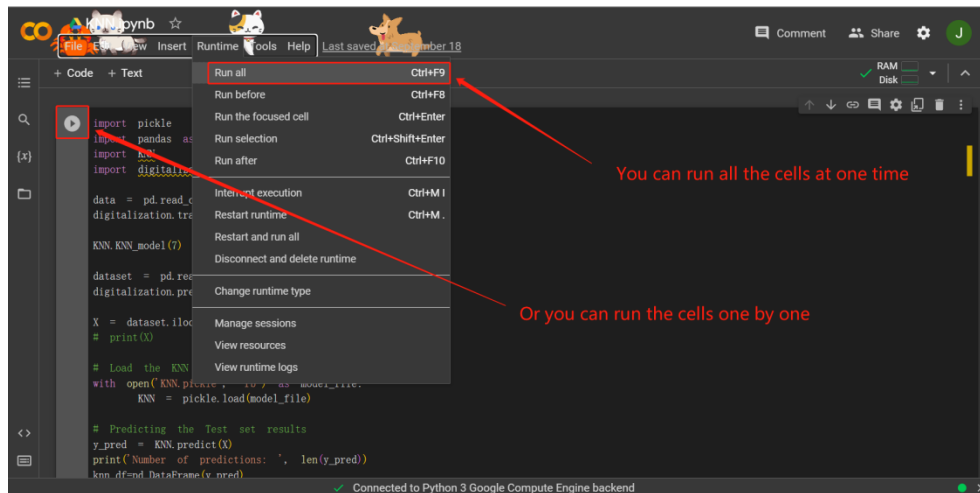


Picture 34: Connect the notebook



Picture 35: Upload our files

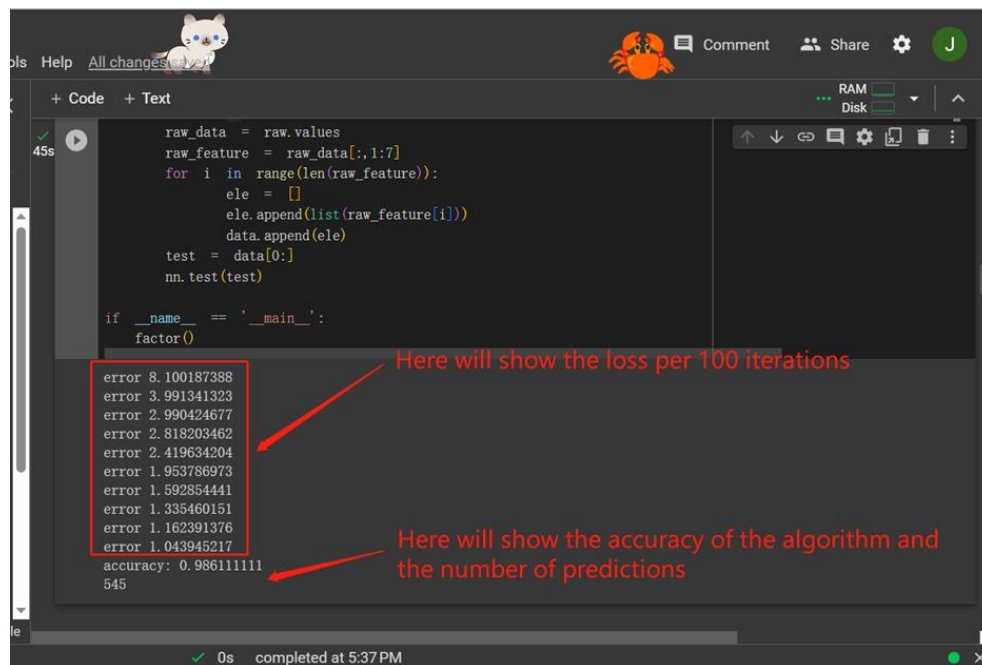
4. Click "Run" button and run the cells one by one or you can run them all in one click. (One by one is recommended)



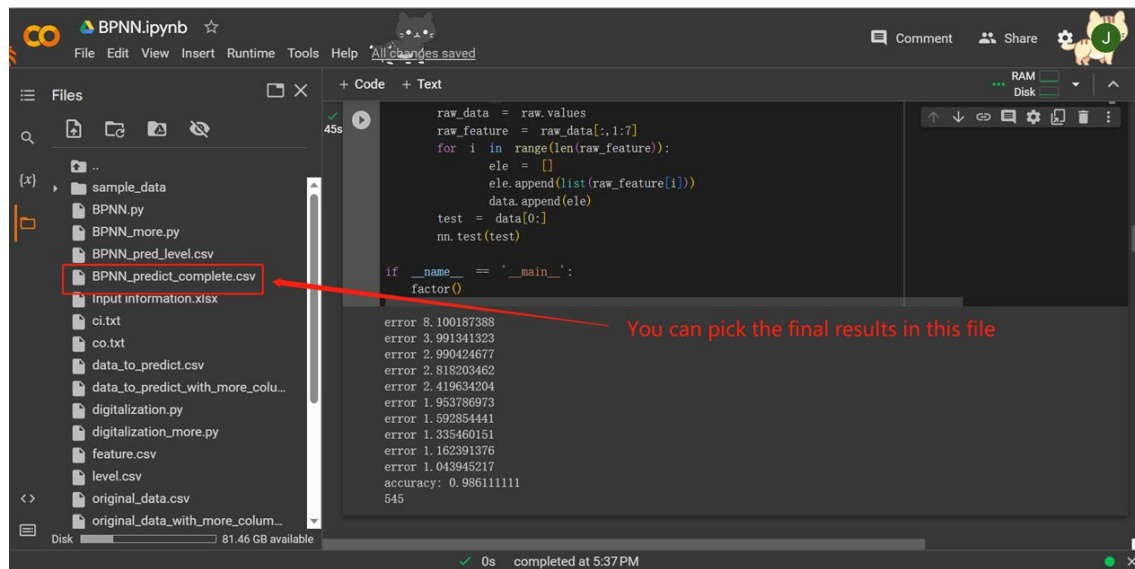
Picture 36: Run the BPNN program

4.2. Check the output of BPNN

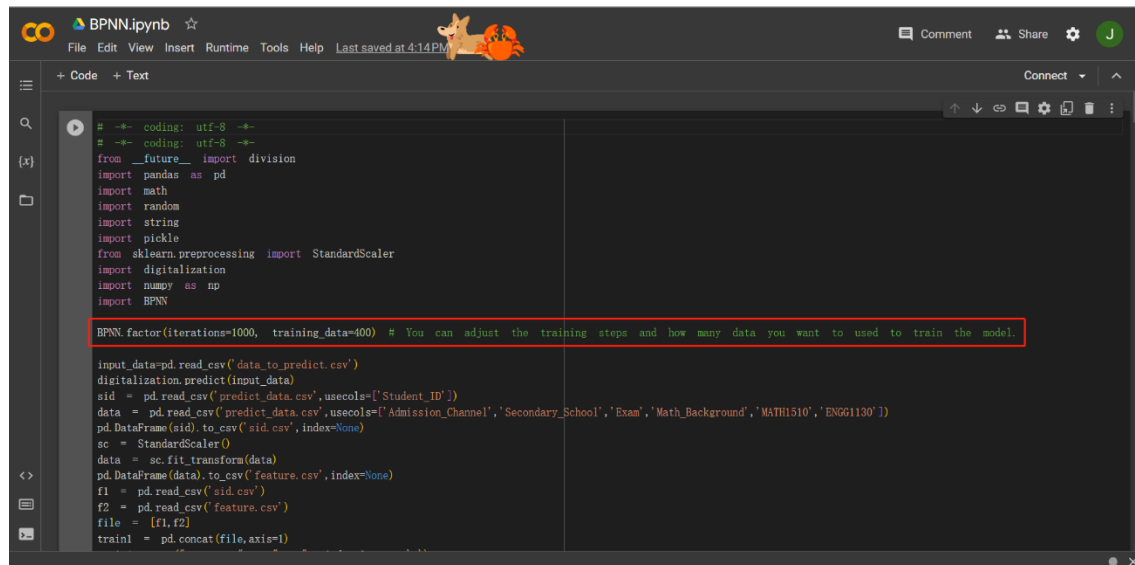
1. Output of the first cell shows the loss and accuracy of our trained BPNN classifier, which also represent the accuracy of final predict result. The predict level are also printed in the output for the users to have a quick check.
2. The final output can be found on the left, named “BPNN_predict_complete.csv”, after opening the file we can find that the predict level is on the rightest column.
3. You can adjust the training iterations and ratio of training data and see their impacts on accuracy.



Picture 37: First output



Picture 38: Outcome



Picture 39: Adjust iterations and ratio of training data

4.3. Want to try your own data?

Follow the steps below:

1. First, open the "Input information.xlsx" from the "BPNN" file we have already downloaded, fill in your personal information. Notice that in "Exam" and "Math_Background" column, only Gaokao and HKDSE are accepted.
2. Upload the file to BPNN program and run the second cell.
3. Then pick up the "Personal_BPNN_predict_complete.csv" file to check the prediction of your grade!

	A	B	C	D	E	F	G	H
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENG1130	ENG2020
2	1	JUPAS	CCC HEEP WOH COLLEGE	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A	A	
3	2	Non-JUPAS (Local)	SUEN WAN PUB HO CHUEN YIU MEMORIAL COL	HKDSE	HKDSE.5+	B+	C	
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								
18								
19								
20								
21								
22								
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26								
27								
28								
29								

Picture 40: Input your information

Upload your "Input information.xlsx" here

Input the number of students you want to predict the level.
In the default form it's "2".

Please input the number of students:

If you want to add some extra columns of student features

Picture 41: Run the second cell

You can check the result in this file.

Or you can just see the predictions here

Please input the number of students: 2
Number of predictions: 2
['A', 'C']

If you want to add some extra columns of student features

Picture 42: Two ways to check the predictions

4.4. Want to add some extra factors?

Follow the steps below:

1. We can take a look at “original_data_with_more_columns.csv” in “BPNN” file.
2. After training the model with the dataset in previous slide, you can make a csv document contains the students’ information you want to predict their level. (“data_to_predict_with_more_columns.csv”)

Original No.

We add two more courses as factors for example

	A	B	C	D	E	F	G	H	I	J
1	Original No.	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
2	17 JUPAS		TSUEN WAN PUB HO CHUEN YIU MEMORIAL COLL	HKDSE	HKDSE:5	B+	B+	A-	B	B+
3	20 JUPAS		CHRISTIAN ALLIANCE S C CHAN MEM COLL	HKDSE	HKDSE:5+	A-	B	B	B+	B
4	25 JUPAS		MUNSANG COLLEGE (HONG KONG ISLAND)	HKDSE	HKDSE:4	B	A-	A-	B+	A
5	29 JUPAS		PO LEUNG KUK LAWS FOUNDATION COLLEGE	HKDSE	HKDSE:4	B+	B-	B+	A-	B-
6	36 JUPAS		STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5	B-	B+	B+	B	B+
7	37 JUPAS		METHODIST CHURCH HK WESLEY COLLEGE	HKDSE	HKDSE:5	D	D+	B+	B+	D
8	50 JUPAS		Others	HKDSE	HKDSE:5	B-	B+	D	D+	C
9	55 JUPAS		KWOK TAK SENG CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:5	B	B	B	C+	B
10	62 JUPAS		YWCA HIOE TJO YOENG COLLEGE	HKDSE	HKDSE:4	B	B+	B-	B	B+
11	82 JUPAS		CCC CHUEN YUEN COLLEGE	HKDSE	HKDSE:5	B	F	B	B+	F
12	88 JUPAS		HKMA K S LO COLLEGE	HKDSE	HKDSE:4	C	C+	C-	F	C
13	92 JUPAS		LIONS COLLEGE	HKDSE	HKDSE:4	B	C-	C+	C+	C
14	94 JUPAS		YCH LIM POR YEN SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	A-	C-	C-	A
15	98 JUPAS		HK SKH BISHOP HALL SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	B+	A-	A-	B+
16	101 JUPAS		CHRISTIAN ALLIANCE S W CHAN MEM COLL	HKDSE	HKDSE:5	B-	B-	B+	B+	B-
17	109 JUPAS		VALTORTA COLLEGE	HKDSE	HKDSE:5**	A-	A	B-	B+	A
18	119 JUPAS		AOG HEBRON SECONDARY SCHOOL	HKDSE	HKDSE:5	B-	A-	A	A	A
19	129 JUPAS		PUI CHING MIDDLE SCHOOL	HKDSE	HKDSE:5**	A-	A	A-	A-	A
20	130 JUPAS		TWGH LI KA SHING COLLEGE	HKDSE	HKDSE:5	C+	C-	A	A	C
21	142 JUPAS		TANG KING PO SCHOOL	HKDSE	HKDSE:4	B+	B-	C-	C+	B-
22	144 JUPAS		PLK MRS MA KAM MING CHEUNG FOOK SIEN COL	HKDSE	HKDSE:4	B+	B-	B-	B-	B-
23	147 JUPAS Appeal-in		STFA YUNG YAU COLLEGE	HKDSE	HKDSE:5	B-	C	B-	B-	C
24	168 JUPAS		FUKIEN SECONDARY SCHOOL	HKDSE	HKDSE:5+	A-	B+	C	C	B+
25	172 JUPAS		SKH LAM WOO MEMORIAL SECONDARY SCHOOL	HKDSE	HKDSE:4	B	C+	B+	B+	C

Picture 43: original_data_with_more_columns.csv

Student_ID

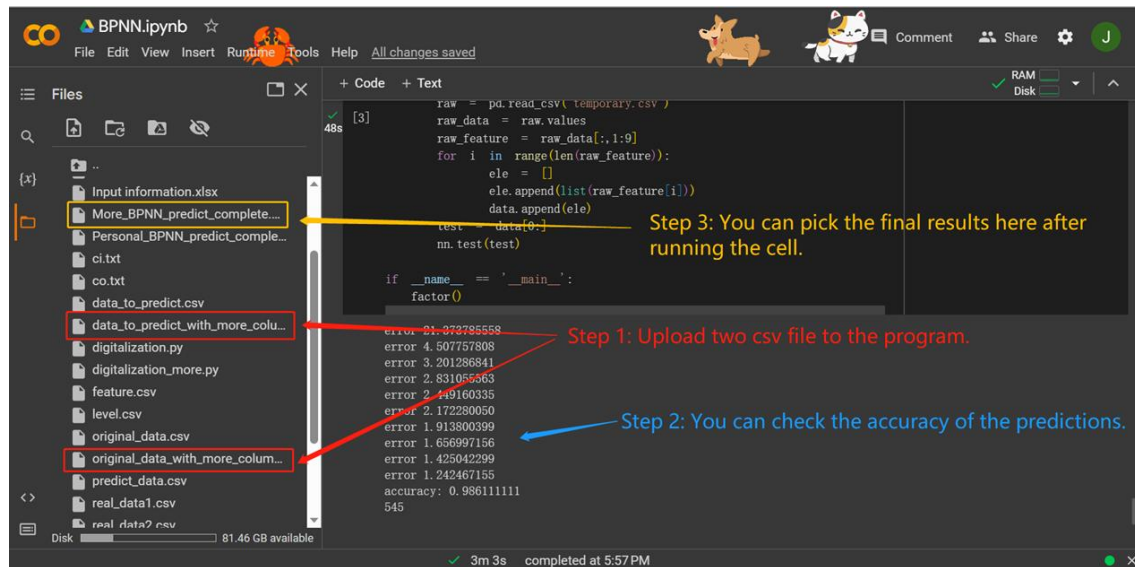
After running the cell, the level of these students will be added

	A	B	C	D	E	F	G	H	I	J
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
2	627 JUPAS		CHAN SUI KI (LA SALLE) COLLEGE	HKDSE	HKDSE:5+	A-	A-	B-	B-	
3	882 JUPAS		CHRISTIAN ALLIANCE CHENG WING GEE COLL	HKDSE	HKDSE:4	D	F	B	B+	
4	801 JUPAS		MA ON SHAN TSUNG TSIN SECONDARY SCHOOL	HKDSE	HKDSE:5+	B-	B	C	C+	
5	1087 Mainland Fee-paying		Others	Mainland	(Mainland Gaokao (JEE):<110	B-	B	B-	C+	
6	224 JUPAS		ELEGANTIA COLLEGE (SPONSORED BY ED CONV)	HKDSE	HKDSE:5+	B-	F	B-	B-	
7	240 JUPAS		SKH BISHOP MOK SAU TSENG SEC SCH	HKDSE	HKDSE:5+	A-	A-	B-	B+	
8	911 JUPAS		Others	HKDSE	HKDSE:5**	A	A	B-	B+	
9	917 JUPAS		DIOCESAN BOYS SCHOOL	HKDSE	HKDSE:4	D	B-	B	B+	
10	624 Mainland		Others	Mainland	(Mainland Gaokao (JEE):120-129	A-	A	B	A	
11	724 JUPAS		SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B	B-	B+	
12	546 JUPAS		KWUN TONG MARYKNOLL COLLEGE	HKDSE	HKDSE:5+	A	A	B	B+	
13	410 Mainland		Others	Mainland	(Mainland Gaokao (JEE):130-139	B+	B	A-	B	
14	492 JUPAS		Others	HKDSE	HKDSE:5	B-	A-	B	B-	
15	986 JUPAS		Others	HKDSE	HKDSE:5	C+	B+	A-	B+	
16	1046 JUPAS		HELEN LIANG MEMORIAL SEC SCH (SHATIN)	HKDSE	HKDSE:5	A-	B+	B	B	
17	272 JUPAS		HEUNG TO MIDDLE SCHOOL	HKDSE	HKDSE:4	B-	C-	A-	A-	
18	803 JUPAS		Others	HKDSE	HKDSE:4	B-	B+	B	B-	
19	383 JUPAS		LAI KING CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:4	C-	B	B-	B	
20	881 JUPAS		YAN OI TONG TIN KA PING SECONDARY SCHOOL	HKDSE	HKDSE:4	B-	C+	B+	B	
21	932 JUPAS		SKH TANG SHIU KIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B+	B-	A-	
22	120 Mainland		Others	Mainland	(Mainland Gaokao (JEE):120-129	A	A	A-	A-	
23	732 JUPAS		HKSVC&JA WONG TAI SHAN MEM COLL	HKDSE	HKDSE:4	C	B	B+	B+	
24	474 JUPAS		SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	A	B-	C-	
25	776 JUPAS		STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5+	A	A	B-	B-	

Picture 44: data_to_predict_with_more_columns.csv

3. Then upload two csv files to the program.

- Run the sixth cell then you can check the accuracy and pick the final results in the “More_BPNN_predict_complete.csv”.

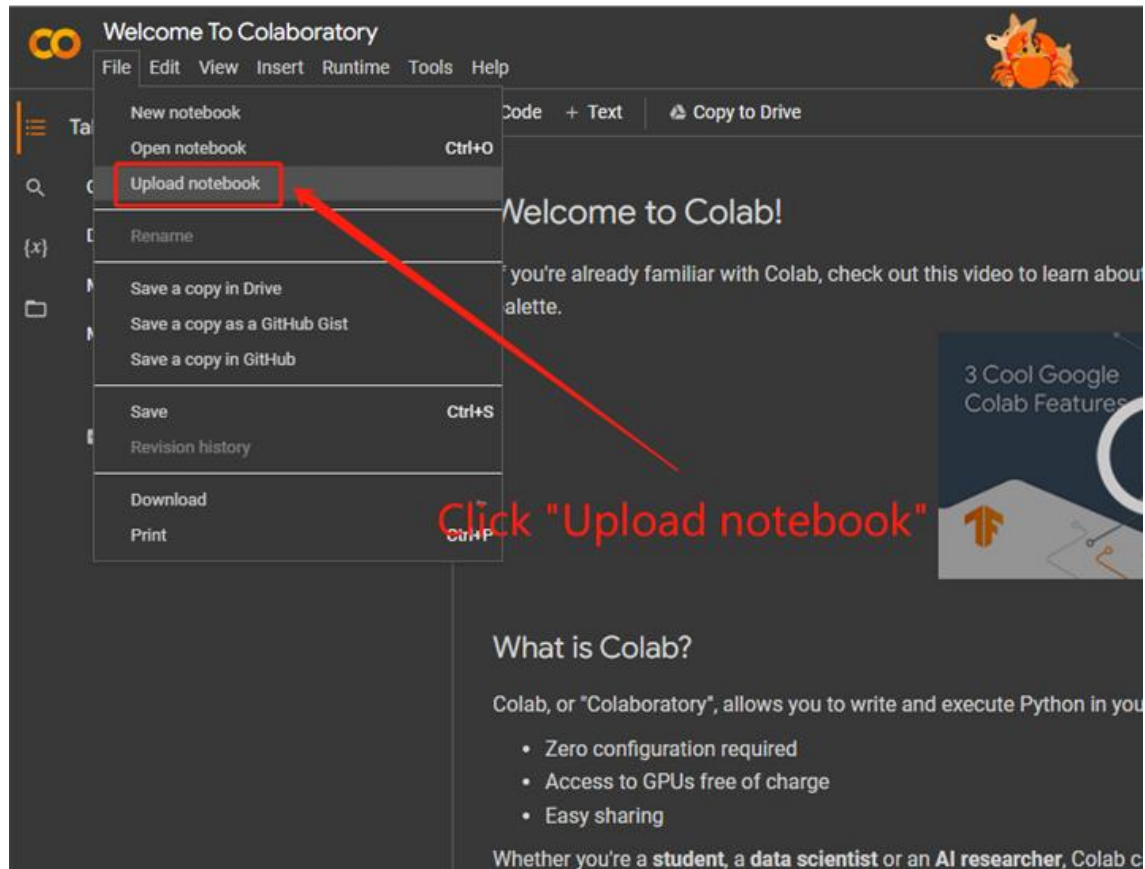


Picture 45: Run the third cell and get your results

5. Try the SVC program

5.1. Preparations

1. Upload the “SVC.ipynb” from the file “Main_code”.

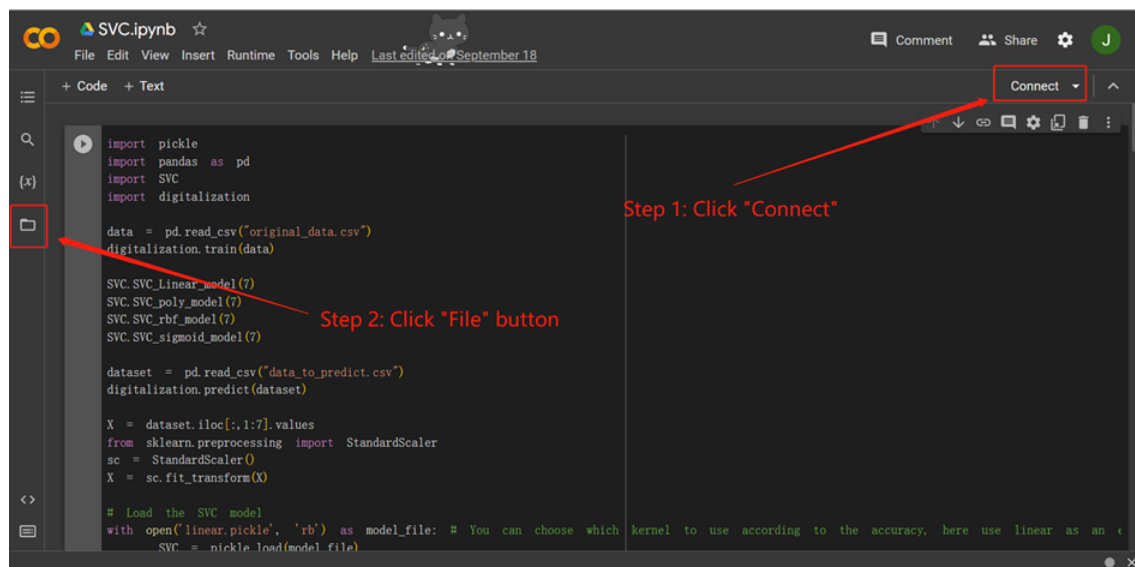


Picture 46: Upload the program

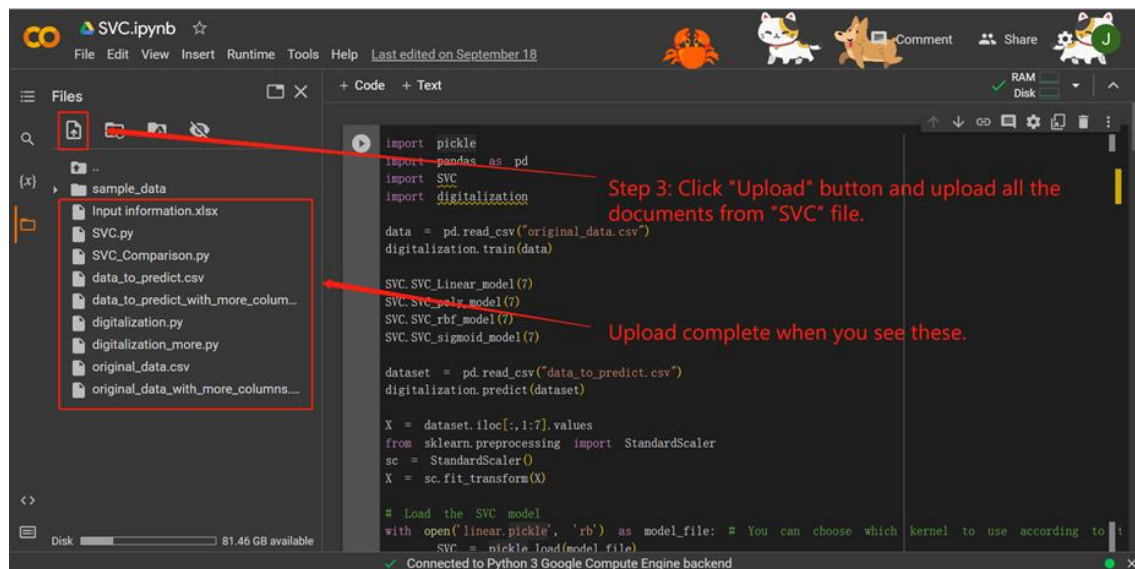


Picture 47: Select SVC.ipynb

2. Connect the notebook to online processor.
3. Upload all the documents from “SVC” file to the notebook.

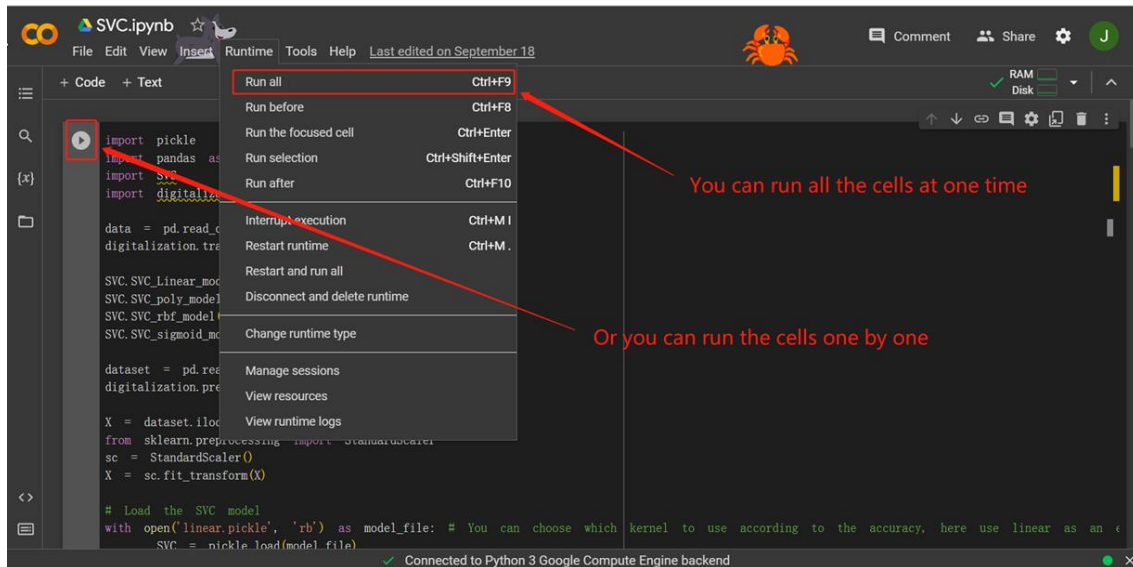


Picture 48: Connect the notebook



Picture 49: Upload our files

4. Click "Run" button and run the cells one by one or you can run them all in one click. (One by one is recommended)



Picture 50: Run the SVC program

5.2. Check the output of SVC

1. Output of the first cell shows the accuracy of our trained SVC classifier, which also represent the accuracy of final predict result. The predict level are also printed in the output for the users to have a quick check.
2. Output of the second cell shows the relationship between different amounts of training dataset and the accuracy.
3. Output of the third cell shows the relationship between different ratios of training & testing data and the accuracy.
4. The final output can be found on the left, named “SVC_predict_complete.csv”, after opening the file we can find that the predict level is on the rightest column.
5. After comparison, you can adjust the ratio of test data like in the picture below. (Since training data is generally the more the better so there’s no need to change)

```

+ Code + Text

# Predicting the Test set results
y_pred = SVC.predict(X)
print('Number of predictions: ', len(y_pred))

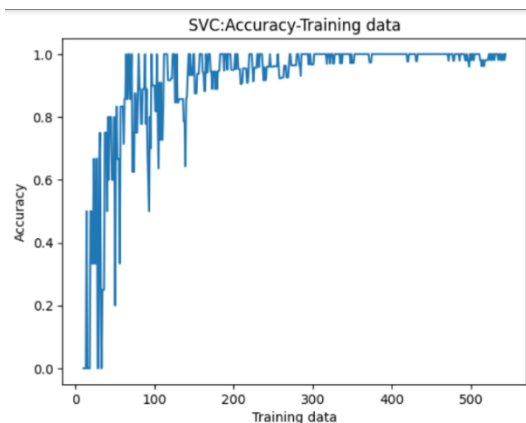
svc_df=pd.DataFrame(y_pred)
svc_df.columns = ['Level']
pd.DataFrame(svc_df).to_csv('SVC_pred_level.csv', index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('SVC_pred_level.csv')
file = [f1,f2]
train = pd.concat(file,axis=1)
train.to_csv("SVC_predict_complete" + ".csv", index=0, sep=',')

Accuracy of the linear kernel Clasification is: 1.0
Accuracy of the poly kernel Clasification is: 0.6333333333333333
Accuracy of the rbf kernel Clasification is: 0.9222222222222223
Accuracy of the sigmoid kernel Clasification is: 0.8222222222222222
Number of predictions: 545

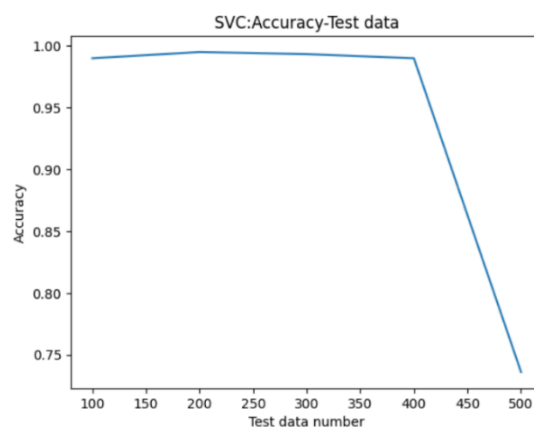
```

Here shows the accuracy of different kernels

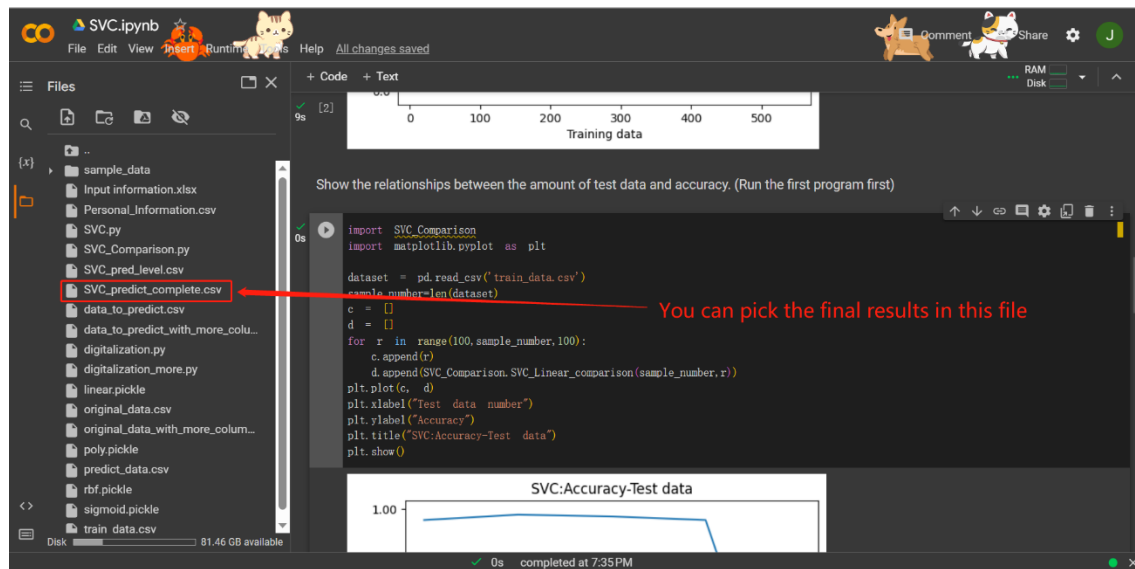
Picture 51: First output



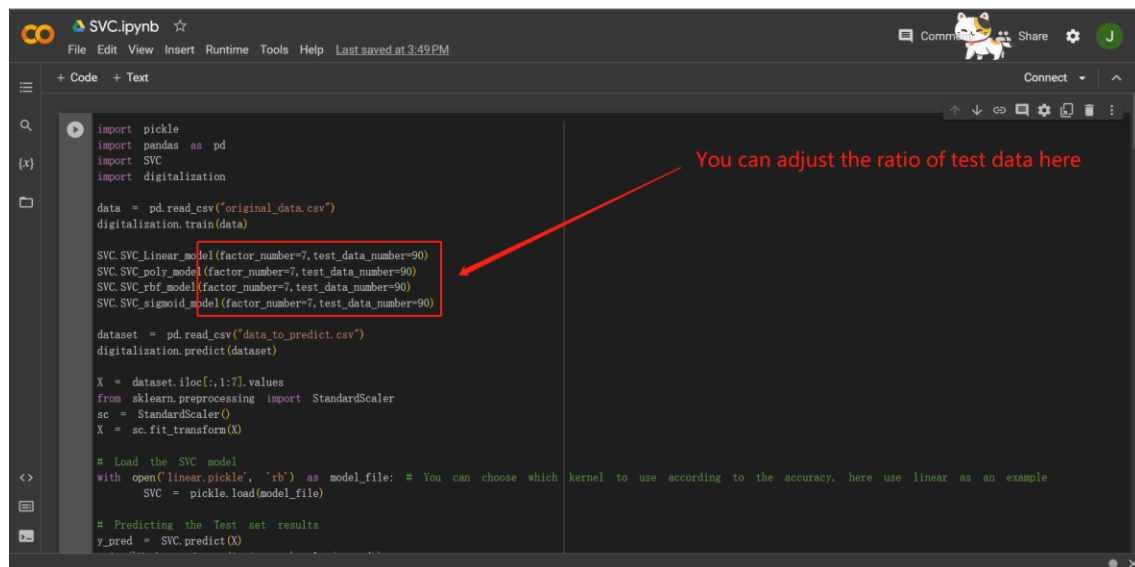
Picture 52: Second output



Picture 53: Third output



Picture 54: Outcome



Picture 55: Adjust the parameters

5.3. Want to try your own data?

Follow the steps below:

1. First, open the "Input information.xlsx" from the "SVC" file we have already downloaded, fill in your personal information. Notice that in "Exam" and "Math_Background" column, only Gaokao and HKDSE are accepted.
2. Upload the file to SVC program and run the fourth cell.
3. You can choose the kernel with higher accuracy to use like in picture 58.

- Then pick up the “Personal_SVC_predict_complete.csv” file to check the prediction of your grade!

A	B	C	D	E	F	G	H
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENG1130
2	1	JUPAS	CCC HEEP WOH COLLEGE	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A	A
3	2	Non-JUPAS (Local)	SUEN WAN PUI HO CHUEN YIU MEMORIAL COL	HKDSE	HKDSE.5+	B+	C
4							
5							
6							
7							
8							
9							
10							
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25							
26							
27							
28							
29							

Picture 56: Input your information

Upload your "Input information.xlsx" here.

Input the number of students you want to predict the level.
In the default form is "2"

```

df1 = df.drop(['School_Information'],axis=1)
pd.DataFrame(df1).to_csv('Personal_Information.csv',index=None)
dataset = pd.read_csv('Personal_Information.csv')
digitalization.predict(dataset)
student_number = int(input('Please input the number of students: '))
X = dataset.iloc[student_number-1:7].values
# print(X)

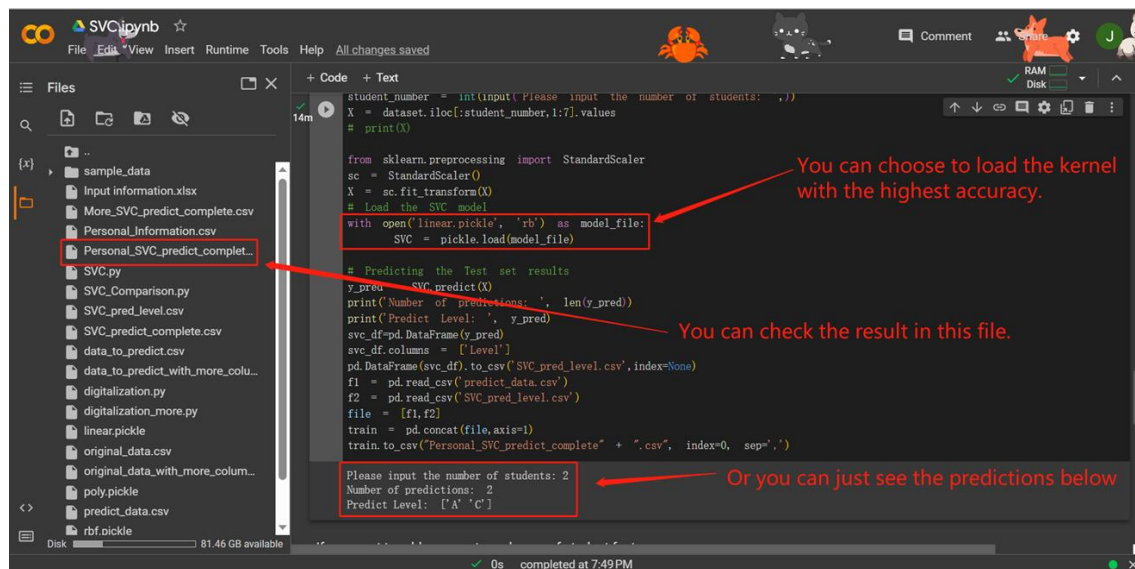
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X = sc.fit_transform(X)
# Load the SVC model
with open('linear.pickle', 'rb') as model_file:
    SVC = pickle.load(model_file)

# Predicting the Test set results
y_pred = SVC.predict(X)
print('Number of predictions: ', len(y_pred))
print('Predict Level: ', y_pred)
svc_df=pd.DataFrame(y_pred)
svc_df.columns = ['Level']
pd.DataFrame(svc_df).to_csv('SVC_pred_level.csv',index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('SVC_pred_level.csv')
file = [f1,f2]
train = pd.concat(file,axis=1)
train.to_csv('Personal_SVC_predict_complete' + '.csv',index=0, sep=',')

```

Please input the number of students:

Picture 57: Run the fourth cell



Picture 58: Two ways to check the predictions

5.4. Want to add some extra factors?

Follow the steps below:

- We can take a look at “original_data_with_more_columns.csv” in “SVC” file.
- After training the model with the dataset in previous slide, you can make a csv document contains the students’ information you want to predict their level. (“data_to_predict_with_more_columns.csv”)

We add two more courses as factors for example

	A	B	C	D	E	F	G	H	I	J
1	Original No.	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
2	17 JUPAS	TSUEN WAN PUB HO CHUEN YIU MEMORIAL COLL	HKDSE	HKDSE:5	B+	B+	A-	B	B+	B+
3	20 JUPAS	CHRISTIAN ALLIANCE S C CHAN MEM COLL	HKDSE	HKDSE:5+	A-	B	B	B+	B	B
4	25 JUPAS	MUNSANG COLLEGE (HONG KONG ISLAND)	HKDSE	HKDSE:4	B	A-	A-	B+	A	A
5	29 JUPAS	PO LEUNG KUK LAWS FOUNDATION COLLEGE	HKDSE	HKDSE:4	B+	B-	B+	A-	B-	B-
6	36 JUPAS	STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5	B-	B+	B+	B	B+	B+
7	37 JUPAS	METHODIST CHURCH HK WESLEY COLLEGE	HKDSE	HKDSE:5	D	D+	B+	B+	D	D
8	50 JUPAS	Others	HKDSE	HKDSE:5	B-	C+	D	D+	C	C
9	55 JUPAS	KWOK TAK SENG CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:5	B	B	B	C+	B	B
10	62 JUPAS	YWCA HIOE TJO YOENG COLLEGE	HKDSE	HKDSE:4	B	B+	B-	B	B+	B+
11	82 JUPAS	CCC CHUEN YUEN COLLEGE	HKDSE	HKDSE:5	B	F	B	B+	F	F
12	88 JUPAS	HKMA K S LO COLLEGE	HKDSE	HKDSE:4	C	C+	C-	F	C	C
13	92 JUPAS	LIONS COLLEGE	HKDSE	HKDSE:4	B	C-	C+	C+	C	C
14	94 JUPAS	YCH LIM POR YEN SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	A-	C-	C-	A	A
15	98 JUPAS	HK SKH BISHOP HALL SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	B+	A-	A-	B+	B+
16	101 JUPAS	CHRISTIAN ALLIANCE S W CHAN MEM COLL	HKDSE	HKDSE:5	B-	B-	B+	B+	B-	B-
17	109 JUPAS	VALTORTA COLLEGE	HKDSE	HKDSE:5**	A-	A	B-	B+	A	A
18	119 JUPAS	AOG HEBRON SECONDARY SCHOOL	HKDSE	HKDSE:5	B-	A-	A	A	A	A
19	129 JUPAS	PUI CHING MIDDLE SCHOOL	HKDSE	HKDSE:5**	A-	A	A-	A-	A	A
20	130 JUPAS	TWGH LI KA SHING COLLEGE	HKDSE	HKDSE:5	C+	C-	A	A	C	C
21	142 JUPAS	TANG KING PO SCHOOL	HKDSE	HKDSE:4	B+	B-	C-	C+	B-	B-
22	144 JUPAS	PLK MRS MA KAM MING CHEUNG FOOK SIEN COL	HKDSE	HKDSE:4	B+	B-	B-	B-	B-	B-
23	147 JUPAS Appeal-in	STFA YUNG YAU COLLEGE	HKDSE	HKDSE:5	B-	C	B-	B-	C	C
24	168 JUPAS	FUKIEN SECONDARY SCHOOL	HKDSE	HKDSE:5+	A-	B+	C	C	B+	B+
25	172 JUPAS	SKH LAM WOO MEMORIAL SECONDARY SCHOOL	HKDSE	HKDSE:4	B	C+	B+	B+	C	C

Picture 59: original_data_with_more_columns.csv

After running the cell, the level of these students will be added

	A	B	C	D	E	F	G	H	I	J
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
2	627 JUPAS		CHAN SUI KI (LA SALLE) COLLEGE	HKDSE	HKDSE-5*	A-	A-	B-	B-	
3	882 JUPAS		CHRISTIAN ALLIANCE CHENG WING GEE COLL	HKDSE	HKDSE-4	D	F	B	B+	
4	801 JUPAS		MA ON SHAN TSUNG TSIN SECONDARY SCHOOL	HKDSE	HKDSE-5*	B-	B	C	C+	
5	1087 Mainland Fee-paying		Others	Mainland (Mainland Gaokao (JEE):<110		B-	B	B-	C+	
6	224 JUPAS		ELEGANTIA COLLEGE (SPONSORED BY ED CONV)	HKDSE	HKDSE-5*	B-	F	B-	B-	
7	240 JUPAS		SKH BISHOP MOK SAU TSENG SEC SCH	HKDSE	HKDSE-5*	A-	A-	B-	B+	
8	911 JUPAS		Others			A	A	B-	B+	
9	917 JUPAS		DIOCESAN BOYS SCHOOL	HKDSE	HKDSE-4	D	B-	B	B+	
10	624 Mainland		Others	Mainland (Mainland Gaokao (JEE):120-129		A-	A	B	A	
11	724 JUPAS		SING YIN SECONDARY SCHOOL	HKDSE	HKDSE-4	B	B	B-	B+	
12	546 JUPAS		KWUN TONG MARYKNOLL COLLEGE	HKDSE	HKDSE-5*	A	A	B	B+	
13	410 Mainland		Others	Mainland (Mainland Gaokao (JEE):130-139		B+	B	A-	B	
14	492 JUPAS		Others	HKDSE	HKDSE-5	B-	A-	B	B-	
15	986 JUPAS		Others	HKDSE	HKDSE-4	C+	B+	A-	B+	
16	1046 JUPAS		HELEN LIANG MEMORIAL SEC SCH (SHATIN)	HKDSE	HKDSE-5	A-	B+	B	B	
17	272 JUPAS		HEUNG TO MIDDLE SCHOOL	HKDSE	HKDSE-4	B-	C-	A-	A-	
18	803 JUPAS		Others	HKDSE	HKDSE-4	B-	B+	B	B-	
19	383 JUPAS		LAI KING CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE-4	C-	B	B-	B	
20	881 JUPAS		YAN OI TONG TIN KA PING SECONDARY SCHOOL	HKDSE	HKDSE-4	B-	C+	B+	B	
21	932 JUPAS		SKH TANG SHIU KIN SECONDARY SCHOOL	HKDSE	HKDSE-4	B	B+	B-	A-	
22	120 Mainland		Others	Mainland (Mainland Gaokao (JEE):120-129		A	A	A-	A	
23	732 JUPAS		HKSYC&JA WONG TAI SHAN MEM COLL	HKDSE	HKDSE-4	C	B	B+	B+	
24	474 JUPAS		SING YIN SECONDARY SCHOOL	HKDSE	HKDSE-4	B	A	B-	C-	
25	776 JUPAS		STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE-5*	A	A	B-	B-	

Picture 60: data_to_predict_with_more_columns.csv

- Then upload two csv files to the program.
- Run the sixth cell then you can check the accuracy and pick the final results in the “More_SVC_predict_complete.csv”.

The screenshot shows a Jupyter Notebook with the following code cells:

```
[5] SVC.SVC_Linear_model(8)

dataset = pd.read_csv("data_to_predict_with_more_columns.csv")
digitalization_more.predict(dataset)

X = dataset.iloc[:,1:6].values
print(X)
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X = sc.fit_transform(X)
# Load the SVC model
with open('linear.pickle', 'rb') as model_file:
    SVC = pickle.load(model_file)

# Predicting the Test set results
y_pred = SVC.predict(X)
print('Number of predictions: ', len(y_pred))
svc_df=pd.DataFrame(y_pred)
svc_df.columns = ['level']
pd.DataFrame(svc_df).to_csv('SVC_pred_level.csv', index=False)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('SVC_pred_level.csv')
file = [f1,f2]
train = pd.concat(file,axis=1)
train.to_csv("More_SVC_predict_complete" + ".csv", index=0, sep=',')

Accuracy of the linear kernel Classification is: 0.9908256880733946
Number of predictions: 545
```

Annotations in the image:

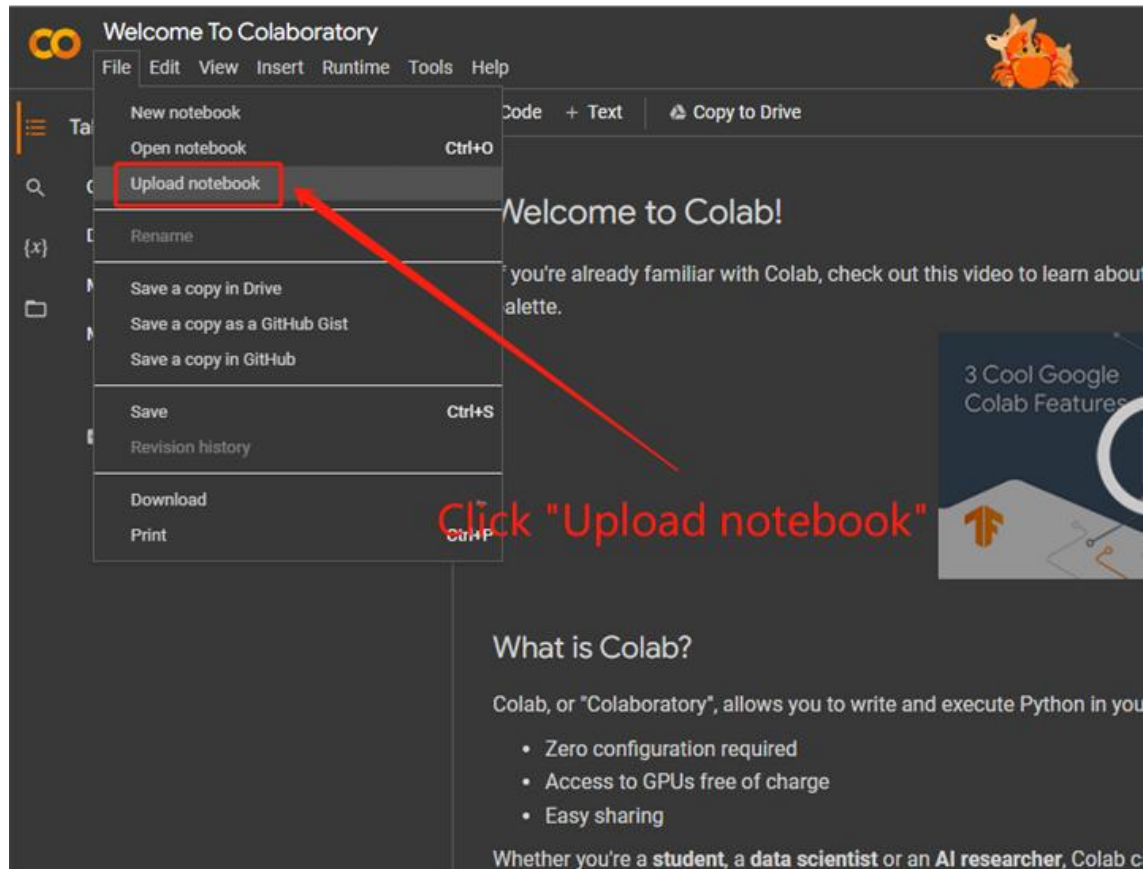
- Step 1: Upload two csv files to the program.** (Points to the file upload area in the Jupyter Notebook interface)
- Step 2: You can check the accuracy of the predictions.** (Points to the accuracy output in the code cell)
- Step 3: You can pick the final results in this csv file after running the cell.** (Points to the 'More_SVC_predict_complete.csv' file in the file explorer)

Picture 61: Run the fifth cell and get your results

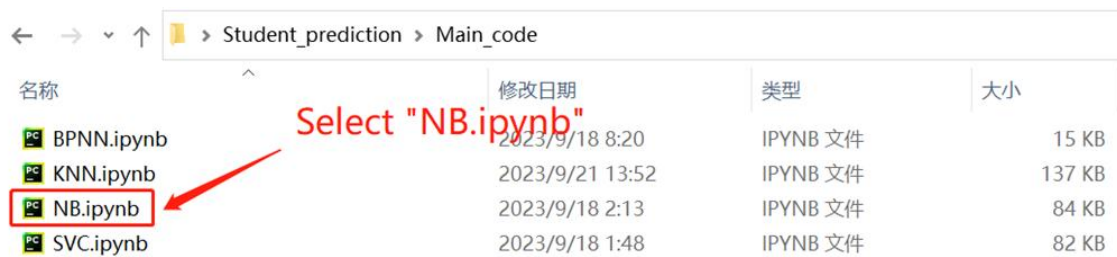
6. Try the NB program

6.1. Preparations

1. Upload the “NB.ipynb” from the file “Main_code”.

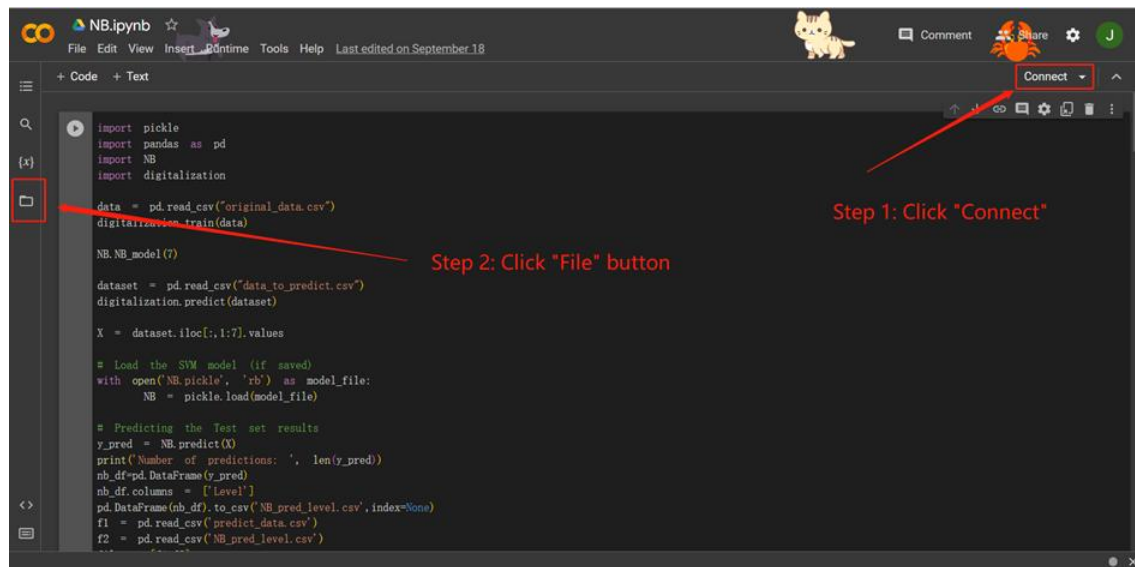


Picture 62: Upload the program

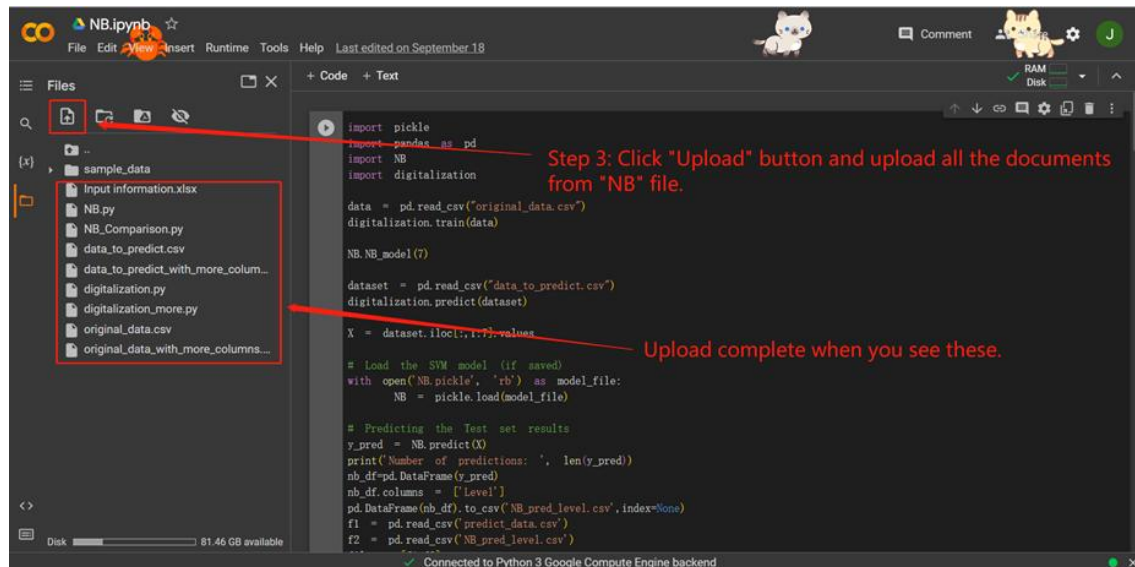


Picture 63: Select NB.ipynb

2. Connect the notebook to online processor.
3. Upload all the documents from “NB” file to the notebook.

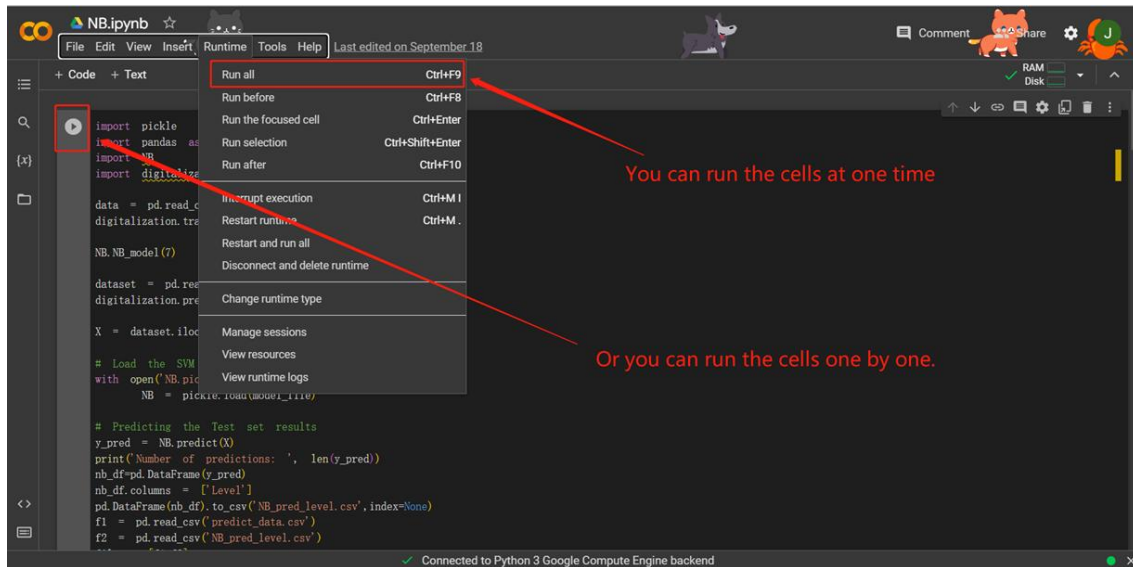


Picture 64: Connect the notebook



Picture 65: Upload our files

4. Click "Run" button and run the cells one by one or you can run them all in one click. (One by one is recommended)



Picture 66: Run the NB program

6.2. Check the output of NB

1. Output of the first cell shows the accuracy of our trained NB classifier, which also represent the accuracy of final predict result. The predict level are also printed in the output for the users to have a quick check.
2. Output of the second cell shows the relationship between different amounts of training dataset and the accuracy.
3. Output of the third cell shows the relationship between different ratios of training & testing data and the accuracy.
4. The final output can be found on the left, named “NB_predict_complete.csv”, after opening the file we can find that the predict level is on the rightest column.
5. After comparison, you can adjust the ratio of test data like in the picture below. (Since training data is generally the more the better so there’s no need to change)

```

+ Code + Text
digitalization.predict(dataset)

X = dataset.iloc[:,1:7].values

# Load the SVM model (if saved)
with open('NB.pickle', 'rb') as model_file:
    NB = pickle.load(model_file)

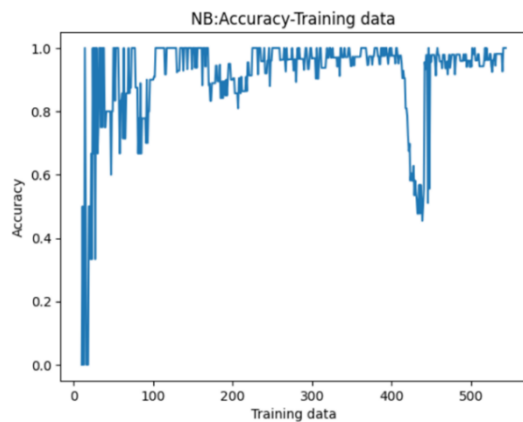
# Predicting the Test set results
y_pred = NB.predict(X)
print('Number of predictions: ', len(y_pred))
nb_df=pd.DataFrame(y_pred)
nb_df.columns = ['Level']
pd.DataFrame(nb_df).to_csv('NB_pred_level.csv', index=None)
f1 = pd.read_csv('predict_data.csv')
f2 = pd.read_csv('NB_pred_level.csv')
file = [f1,f2]
train = pd.concat(file,axis=1)
train.to_csv("NB_predict_complete" + ".csv", index=0, sep=',')

Accuracy of the Naive Bayes Clasification is: 0.9888888888888889
Number of predictions: 545

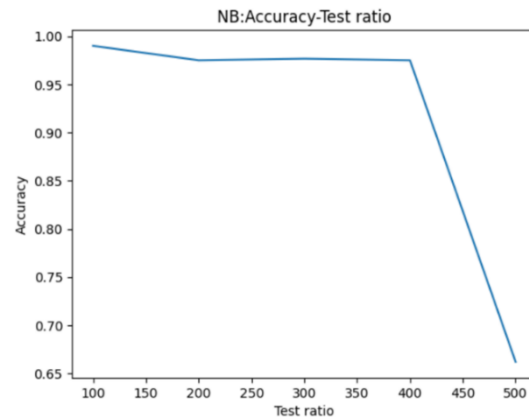
```

Here shows the accuracy and number of predictions

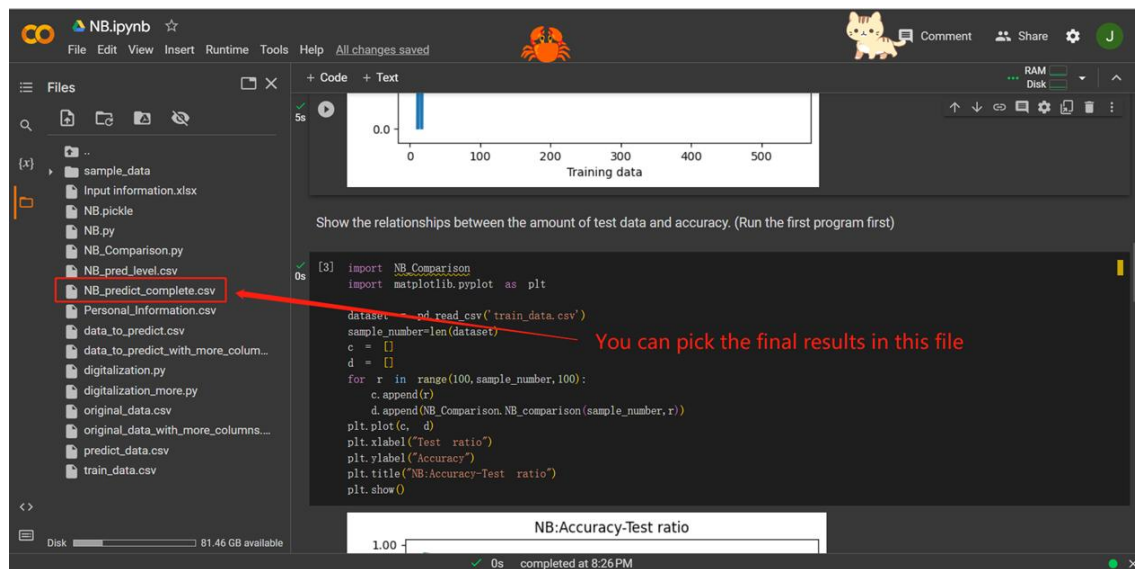
Picture 67: First output



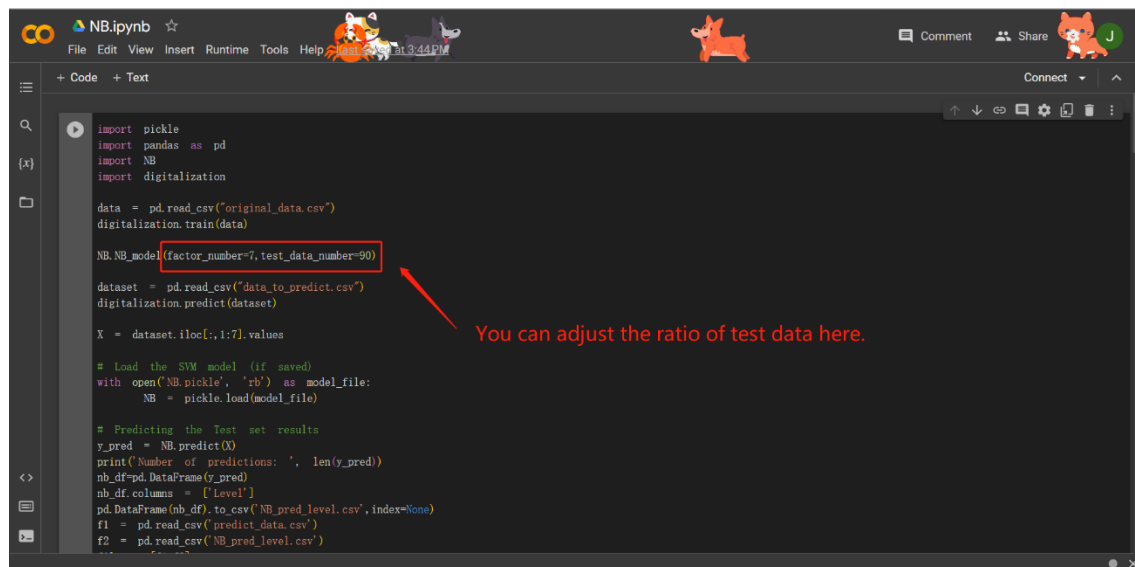
Picture 68: Second output



Picture 69: Third output



Picture 70: Outcome



Picture 71: Adjust the parameters

6.3. Want to try your own data?

Follow the steps below:

1. First, open the “Input information.xlsx” from the “NB” file we have already downloaded, fill in your personal information. Notice that in “Exam” and “Math_Background” column, only Gaokao and HKDSE are accepted.
2. Upload the file to NB program and run the fourth cell.
3. Then pick up the “Personal_NB_predict_complete.csv” file to check the prediction of your grade!

	A	B	C	D	E	F	G	H
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	ENGG2020
2	1	JUPAS	CCC HEEP WOH COLLEGE	Mainland Gaokao (JEE)	Mainland Gaokao (JEE)>140	A	A	
3	2	Non-JUPAS (Local)	SUEN WAN PUB HO CHUEN YIU MEMORIAL COL	HKDSE	HKDSE5+	B+	C	
4								
5								
6								
7								
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29								

Picture 72: Input your information

Upload the "Input information" file first.

Input the number of students you want to predict the level.

In the default form it's "2".

Please input the number of students:

Picture 73: Run the fourth cell

You can check the result in this file.

Or you can just see the predictions here.

Please input the number of students: 2
Number of predictions: 2
Predict Level: ['A' 'A']

Picture 74: Two ways to check the predictions

6.4. Want to add some extra factors?

Follow the steps below:

1. We can take a look at “original_data_with_more_columns.csv” in “NB” file.
2. After training the model with the dataset in previous slide, you can make a csv document contains the students’ information you want to predict their level. (“data_to_predict_with_more_columns.csv”)

A1

Original No.

We add two more courses as factors for example

	A	B	C	D	E	F	G	H	I	J
1	Original No.	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
2	17 JUPAS		TSUEN WAN PUB HO CHUEN YIU MEMORIAL COLL	HKDSE	HKDSE:5	B+	B+	A-	B	B+
3	20 JUPAS		CHRISTIAN ALLIANCE S C CHAN MEM COLL	HKDSE	HKDSE:5+	A-	B	B	B+	B
4	25 JUPAS		MUNSANG COLLEGE (HONG KONG ISLAND)	HKDSE	HKDSE:4	B	A-	A-	B+	A
5	29 JUPAS		PO LEUNG KUK LAWS FOUNDATION COLLEGE	HKDSE	HKDSE:4	B+	B-	B+	A-	B-
6	36 JUPAS		STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5	B-	B+	B+	B	B+
7	37 JUPAS		METHODIST CHURCH HK WESLEY COLLEGE	HKDSE	HKDSE:5	D	D+	B+	B+	D
8	50 JUPAS		Others	HKDSE	HKDSE:5	B-	B+	D	D+	C
9	55 JUPAS		KWOK TAK SENG CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:5	B	B	B	C+	B
10	62 JUPAS		YWCA HIOE TJO YOENG COLLEGE	HKDSE	HKDSE:4	B	B+	B-	B	B+
11	82 JUPAS		CCC CHUEN YUEN COLLEGE	HKDSE	HKDSE:5	B	F	B	B+	F
12	88 JUPAS		HKMA K S LO COLLEGE	HKDSE	HKDSE:4	C	C+	C-	F	C
13	92 JUPAS		LIONS COLLEGE	HKDSE	HKDSE:4	B	C-	C+	C+	C
14	94 JUPAS		YCH LIM POR YEN SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	A-	C-	C-	A
15	98 JUPAS		HK SKH BISHOP HALL SECONDARY SCHOOL	HKDSE	HKDSE:5**	B+	B+	A-	A-	B+
16	101 JUPAS		CHRISTIAN ALLIANCE S W CHAN MEM COLL	HKDSE	HKDSE:5	B-	B-	B+	B+	B-
17	109 JUPAS		VALTORTA COLLEGE	HKDSE	HKDSE:5**	A-	A	B-	B+	A
18	119 JUPAS		AOG HEBRON SECONDARY SCHOOL	HKDSE	HKDSE:5	B-	A-	A	A	A
19	129 JUPAS		PUI CHING MIDDLE SCHOOL	HKDSE	HKDSE:5**	A-	A	A-	A-	A
20	130 JUPAS		TWGH LI KA SHING COLLEGE	HKDSE	HKDSE:5	C+	C-	A	A	C
21	142 JUPAS		TANG KING PO SCHOOL	HKDSE	HKDSE:4	B+	B-	C-	C+	B-
22	144 JUPAS		PLK MRS MA KAM MING CHEUNG FOOK SIEN COL	HKDSE	HKDSE:4	B+	B-	B-	B-	B-
23	147 JUPAS Appeal-in		STFA YUNG YAU COLLEGE	HKDSE	HKDSE:5	B-	C	B-	B-	C
24	168 JUPAS		FUKIEN SECONDARY SCHOOL	HKDSE	HKDSE:5+	A-	B+	C	C	B+
25	172 JUPAS		SKH LAM WOO MEMORIAL SECONDARY SCHOOL	HKDSE	HKDSE:4	B	C+	B+	B+	C

Picture 75: original_data_with_more_columns.csv

A1

Student_ID

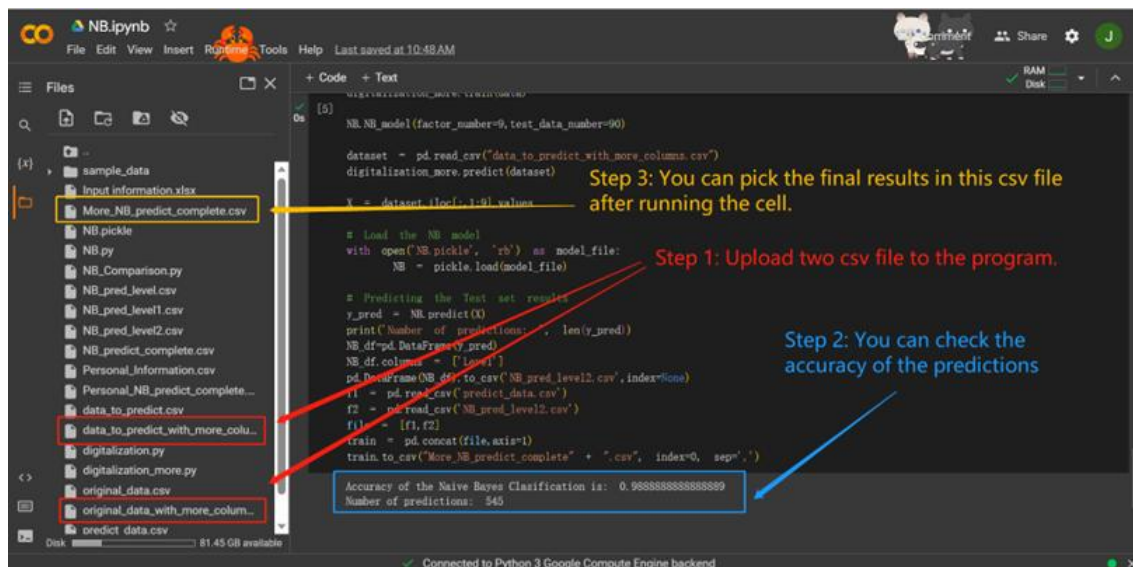
After running the cell, the level of these students will be added

	A	B	C	D	E	F	G	H	I	J
1	Student_ID	Admission_Channel	Secondary_School	Exam	Math_Background	MATH1510	ENGG1130	MATH1010	ENGG1120	ENGG2020
2	627 JUPAS		CHAN SUI KI (LA SALLE) COLLEGE	HKDSE	HKDSE:5+	A-	A-	B-	B-	
3	882 JUPAS		CHRISTIAN ALLIANCE CHENG WING GEE COLL	HKDSE	HKDSE:4	D	F	B	B+	
4	801 JUPAS		MA ON SHAN TSUNG TSIN SECONDARY SCHOOL	HKDSE	HKDSE:5+	B-	B	C	C+	
5	1087 Mainland Fee-paying		Others	Mainland	(Mainland Gaokao (JEE):<110	B-	B	B-	C+	
6	224 JUPAS		ELEGANTIA COLLEGE (SPONSORED BY ED CONV)	HKDSE	HKDSE:5+	B-	F	B-	B-	
7	240 JUPAS		SKH BISHOP MOK SAU TSENG SEC SCH	HKDSE	HKDSE:5+	A-	A-	B-	B+	
8	911 JUPAS		Others	HKDSE	HKDSE:5**	A	A	B-	B+	
9	917 JUPAS		DIOCESAN BOYS SCHOOL	HKDSE	HKDSE:4	D	B-	B	B+	
10	624 Mainland		Others	Mainland	(Mainland Gaokao (JEE):120-129	A-	A	B	A	
11	724 JUPAS		SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B	B-	B+	
12	546 JUPAS		KWUN TONG MARYKNOLL COLLEGE	HKDSE	HKDSE:5+	A	A	B	B+	
13	410 Mainland		Others	Mainland	(Mainland Gaokao (JEE):130-139	B+	B	A-	B	
14	492 JUPAS		Others	HKDSE	HKDSE:5	B-	A-	B	B-	
15	986 JUPAS		Others	HKDSE	HKDSE:5	C+	B+	A-	B+	
16	1046 JUPAS		HELEN LIANG MEMORIAL SEC SCH (SHATIN)	HKDSE	HKDSE:5	A-	B+	B	B	
17	272 JUPAS		HEUNG TO MIDDLE SCHOOL	HKDSE	HKDSE:4	B-	C-	A-	A-	
18	803 JUPAS		Others	HKDSE	HKDSE:4	B-	B+	B	B-	
19	383 JUPAS		LAI KING CATHOLIC SECONDARY SCHOOL	HKDSE	HKDSE:4	C-	B	B-	B	
20	881 JUPAS		YAN OI TONG TIN KA PING SECONDARY SCHOOL	HKDSE	HKDSE:4	B-	C+	B+	B	
21	932 JUPAS		SKH TANG SHIU KIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	B+	B-	A-	
22	120 Mainland		Others	Mainland	(Mainland Gaokao (JEE):120-129	A	A	A-	A-	
23	732 JUPAS		HKSVC&JA WONG TAI SHAN MEM COLL	HKDSE	HKDSE:4	C	B	B+	B+	
24	474 JUPAS		SING YIN SECONDARY SCHOOL	HKDSE	HKDSE:4	B	A	B-	C-	
25	776 JUPAS		STFA LEUNG KAU KUI COLLEGE	HKDSE	HKDSE:5+	A	A	B-	B-	

Picture 76: data_to_predict_with_more_columns.csv

3. Then upload two csv files to the program.

4. Run the fifth cell then you can check the accuracy and pick the final results in the “More_NB_predict_complete.csv”.

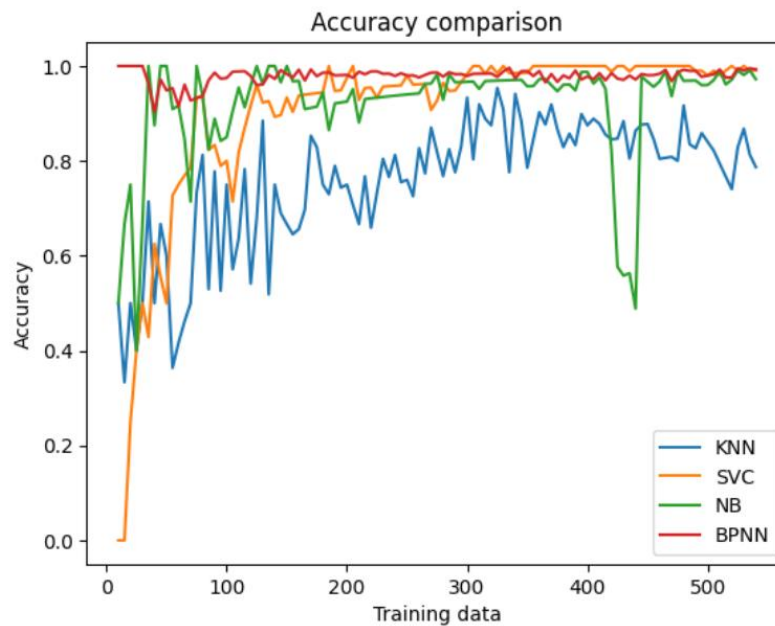


Picture 77: Run the fifth cell and get your results

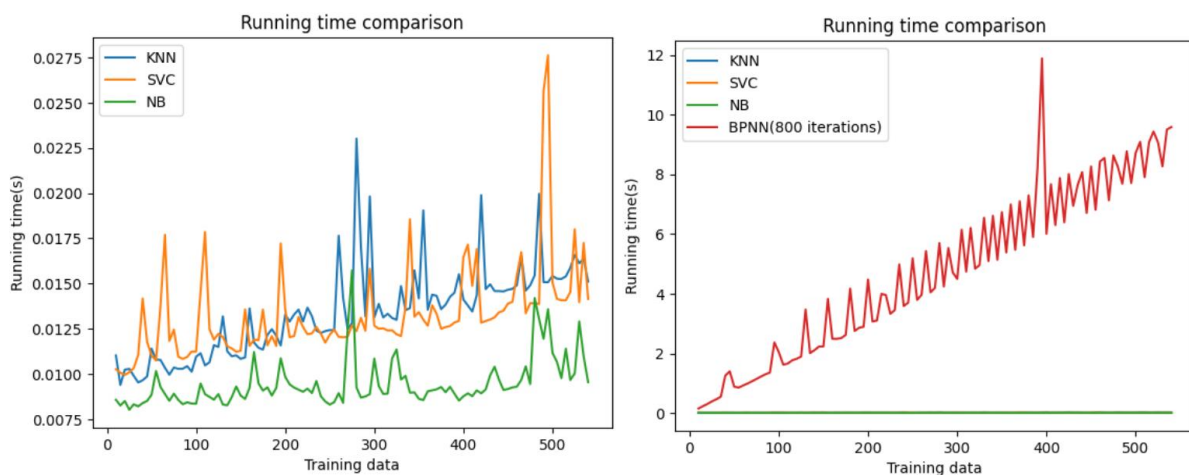
7. Choose the most suitable algorithm

Basically, we choose the algorithm of the highest accuracy to make predictions. However, it is worthy considering the trade-off between the accuracy and computational time.

For example, BPNN will take relatively long time to complete the prediction, but it can handle a more complex task (such as the students' data are very discrete). Therefore, in some simple cases traditional machine learning may have a higher efficiency.



Picture 78: Comparison of four algorithms in accuracy



Picture 79: Comparison of four algorithms in computational time